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**Nelson et al.**

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(54) **DISPLAY OF A VIRTUAL PLAYER IN  
MULTIPLE VIRTUAL REALITY  
ENVIRONMENTS**

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(2013.01); **G07F 17/3223** (2013.01)

(58) **Field of Classification Search**  
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17/3223

See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

2004/0009812 A1\* 1/2004 Scott ..... G07F 17/3251  
463/28  
2011/0201414 A1\* 8/2011 Barclay ..... G07F 17/3206  
463/25

2014/0235348 A1\* 8/2014 Liang ..... A63B 60/46  
463/36  
2015/0221183 A1\* 8/2015 Colvin ..... G07F 17/3211  
463/11  
2020/0043233 A1\* 2/2020 Nelson ..... G06F 3/011

#### OTHER PUBLICATIONS

PSA: Many of the realistic skins on the market are marked “fictional”, Mar. 29, 2018, [https://www.reddit.com/r/Warthunder/comments/883kp9/psa\\_many\\_of\\_the\\_realistic\\_skins\\_on\\_the\\_market\\_are/](https://www.reddit.com/r/Warthunder/comments/883kp9/psa_many_of_the_realistic_skins_on_the_market_are/) (Year:2018) (Year: 2018).\*  
War Thunder (Year: 2018).\*

\* cited by examiner

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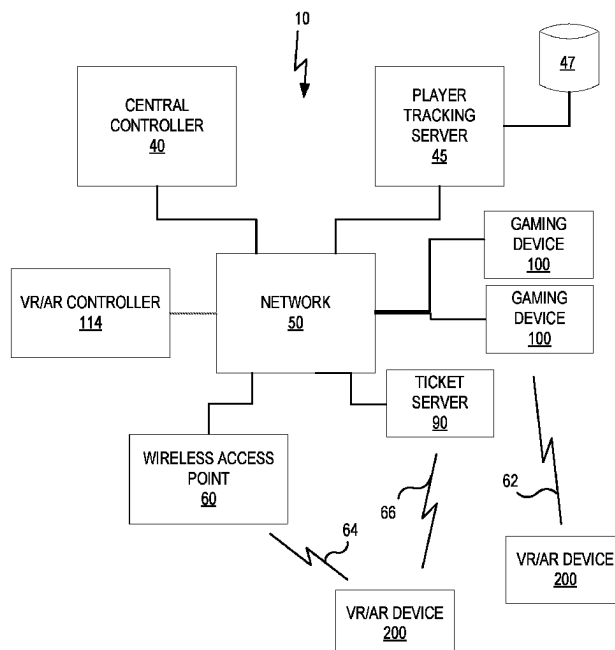
Assistant Examiner — Shauna-Kay Hall

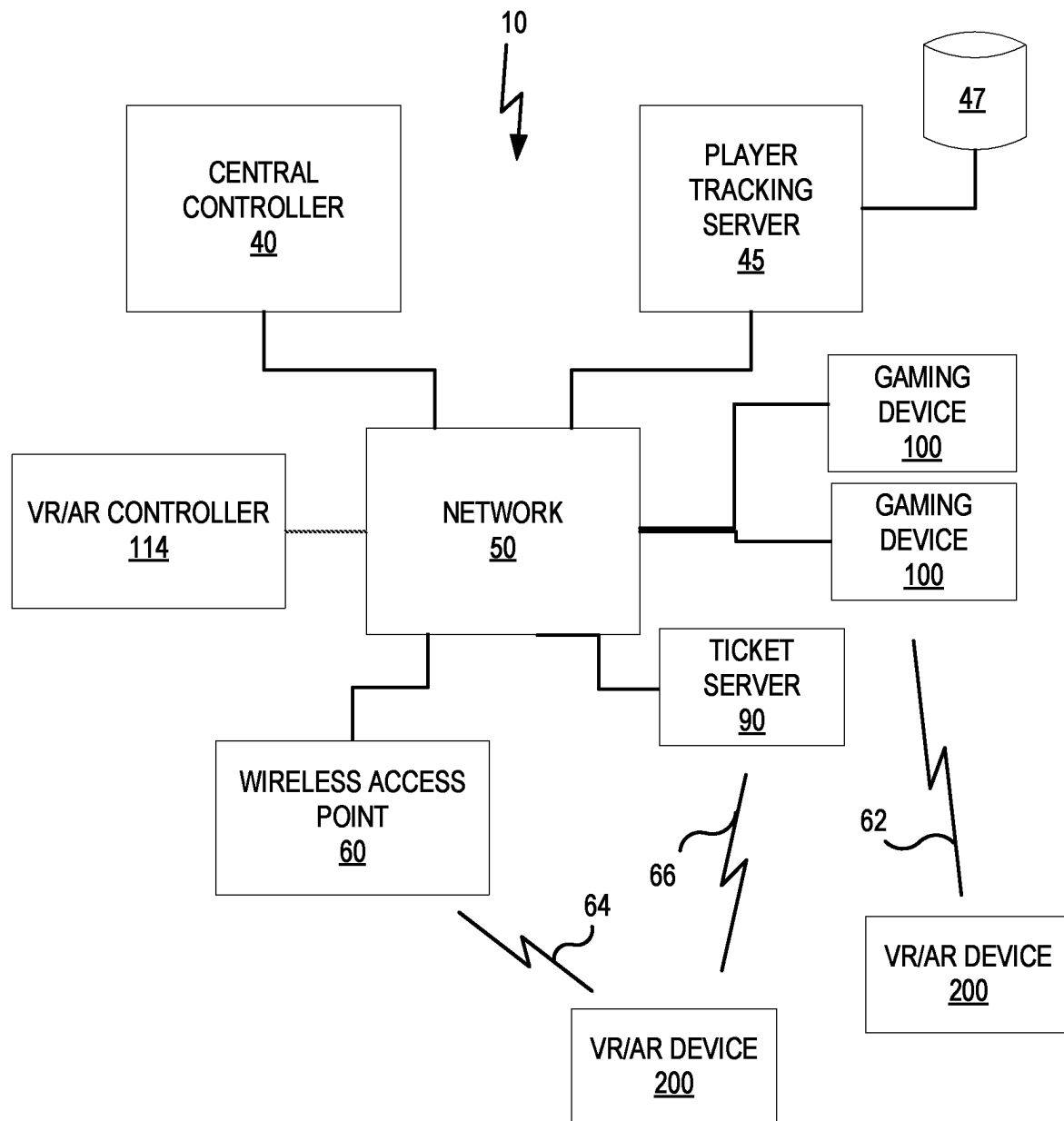
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(57) **ABSTRACT**

A real-time environmental model of a plurality of shared virtual environments (SVEs) is provided, each SVE including a plurality of virtual persons and a plurality of virtual gaming devices. First display data corresponding to a first SVE is transmitted to a first player device, including user display data to render a portion of the first SVE based on a virtual orientation of the first player device and a virtual location of the first player in the first SVE. Second display data corresponding to a second SVE is transmitted to a second player device, including user display data that causes a display device in the second player device to render a portion of the second SVE, including a virtual person associated with the first player, based on a virtual orientation of the second player device and a virtual location of the second player in the second SVE.

**18 Claims, 18 Drawing Sheets**



**FIG. 1**

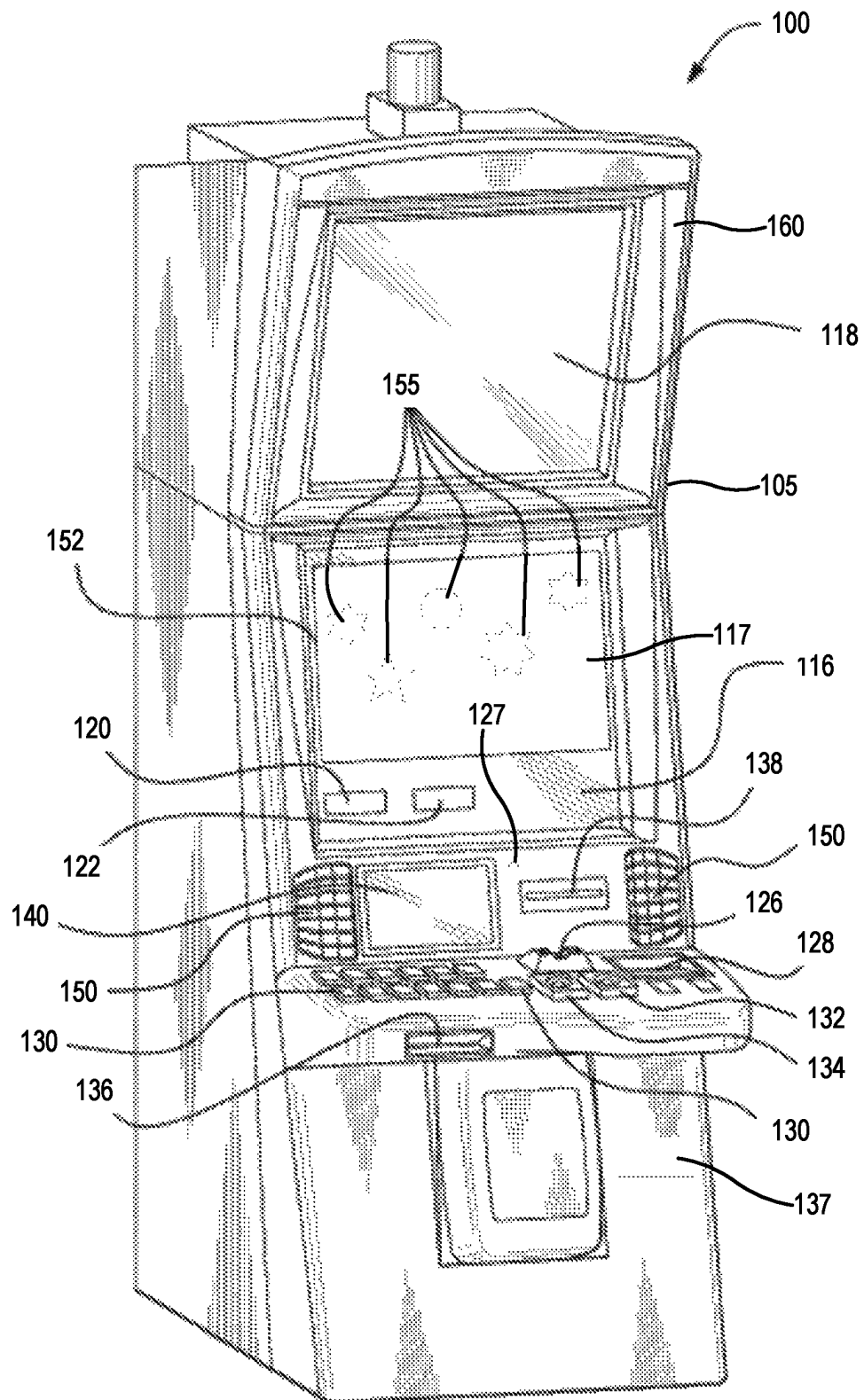
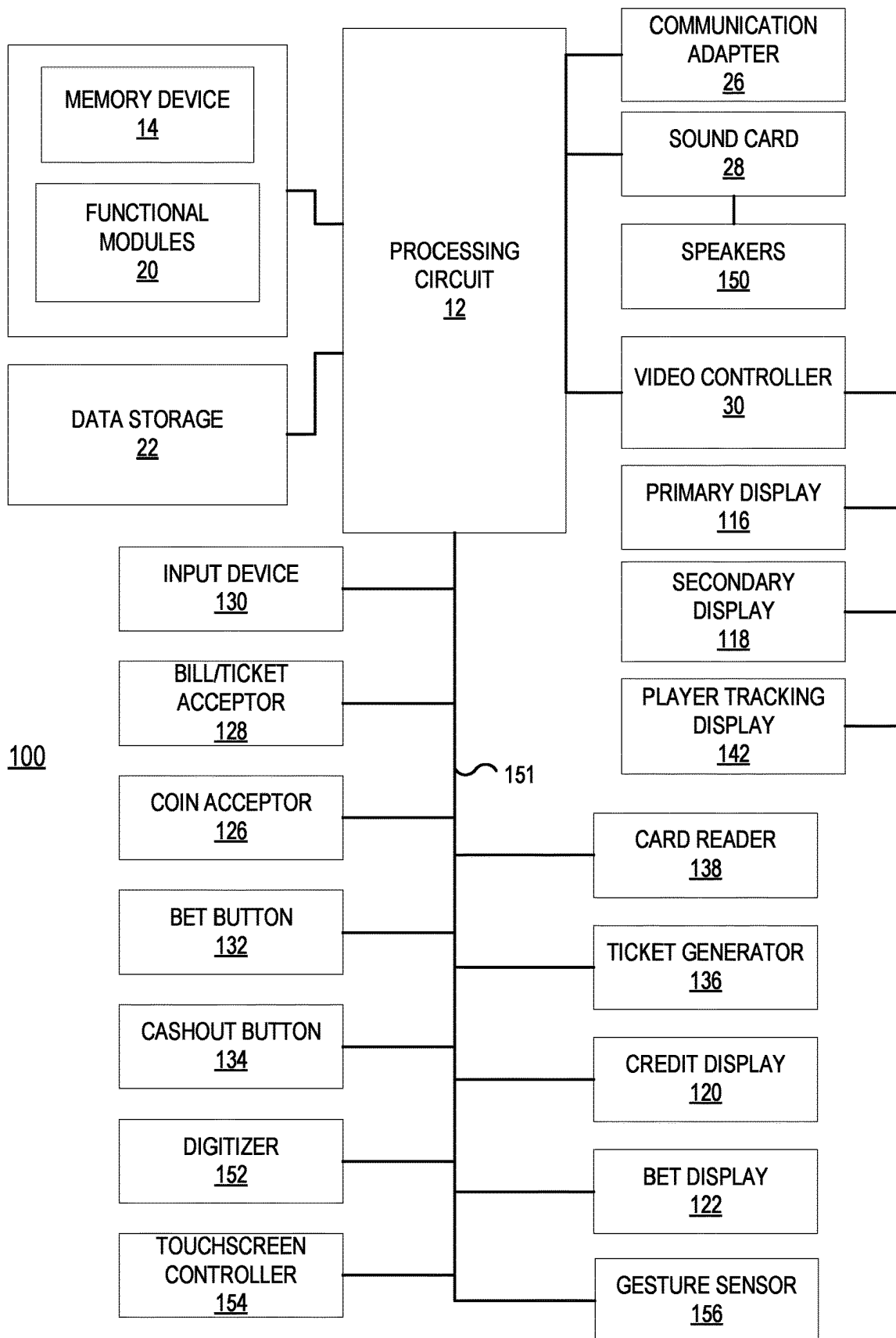
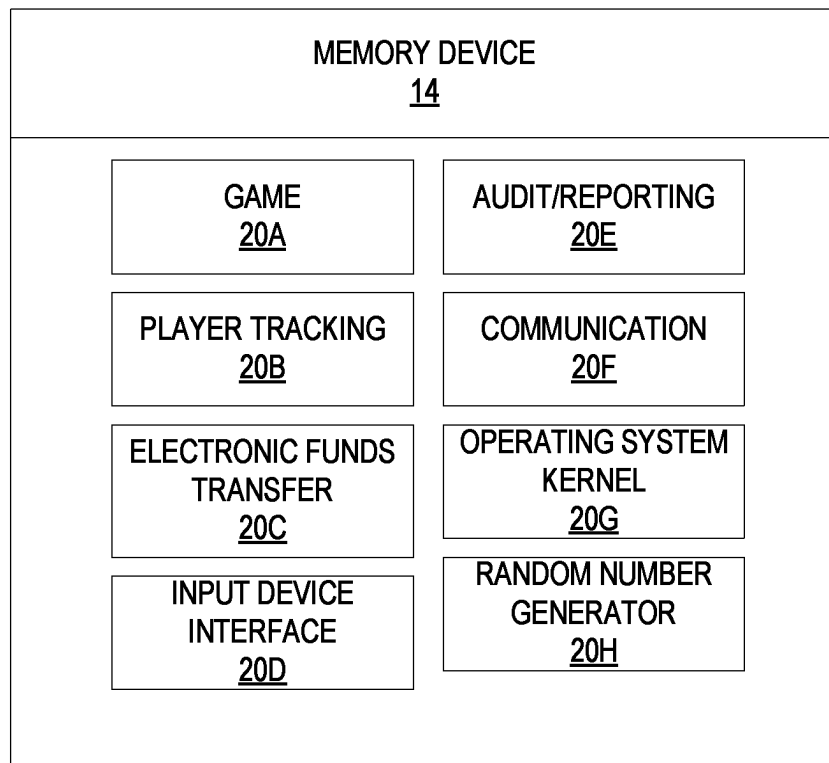
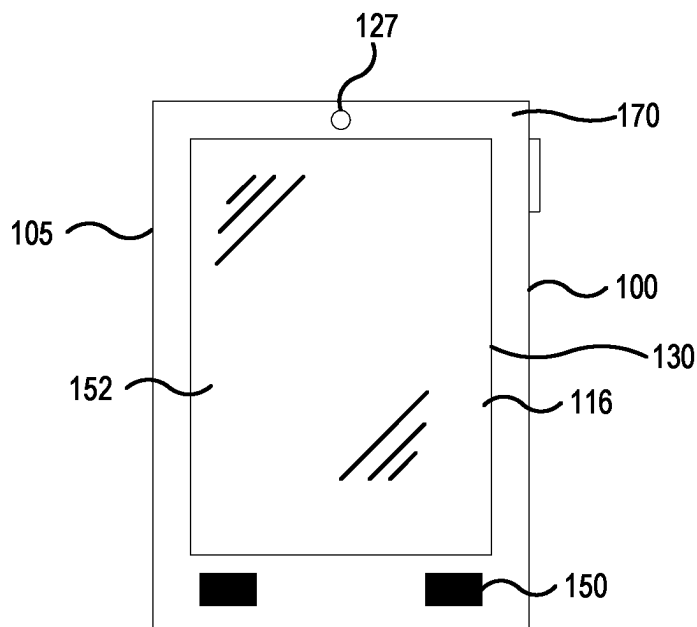


FIG. 2A

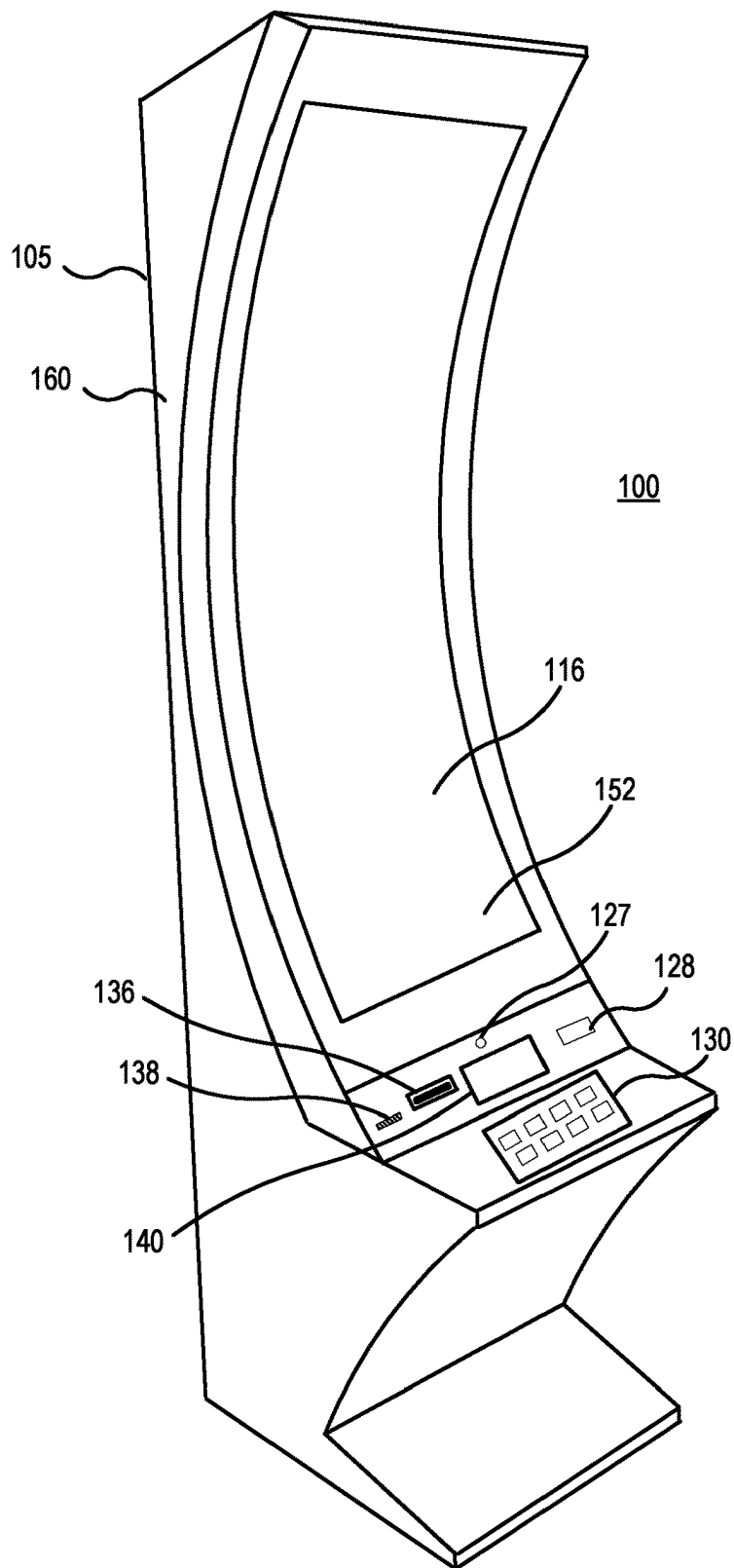
**FIG. 2B**



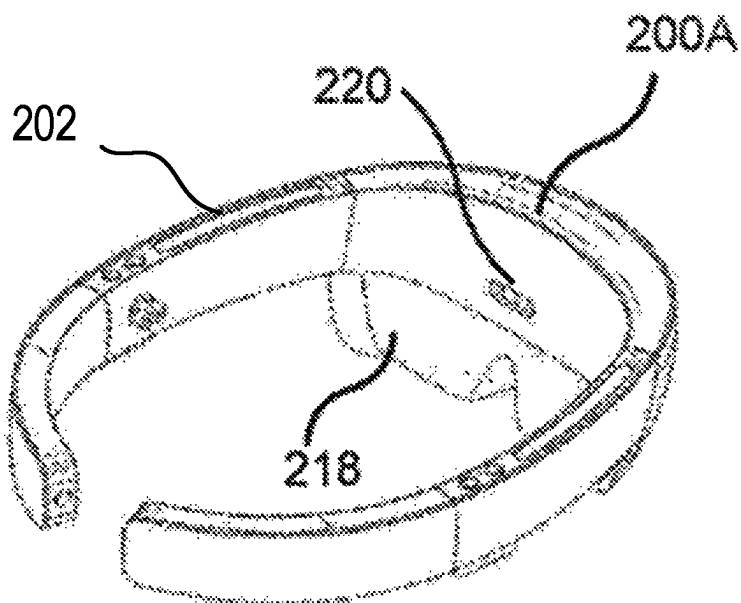
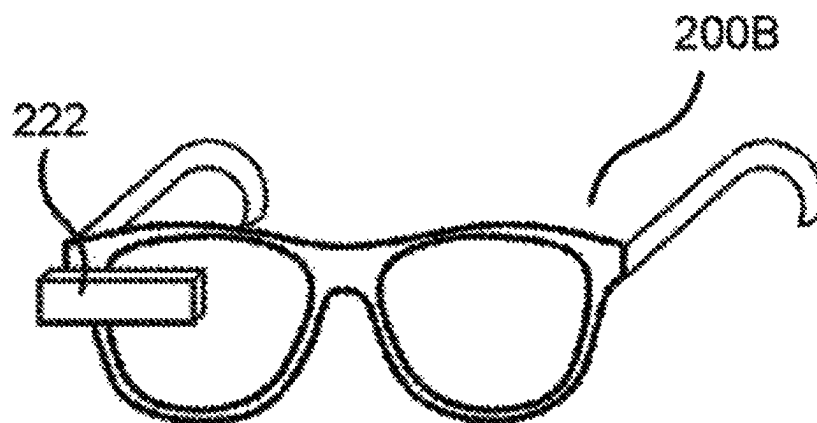
**FIG. 2C**

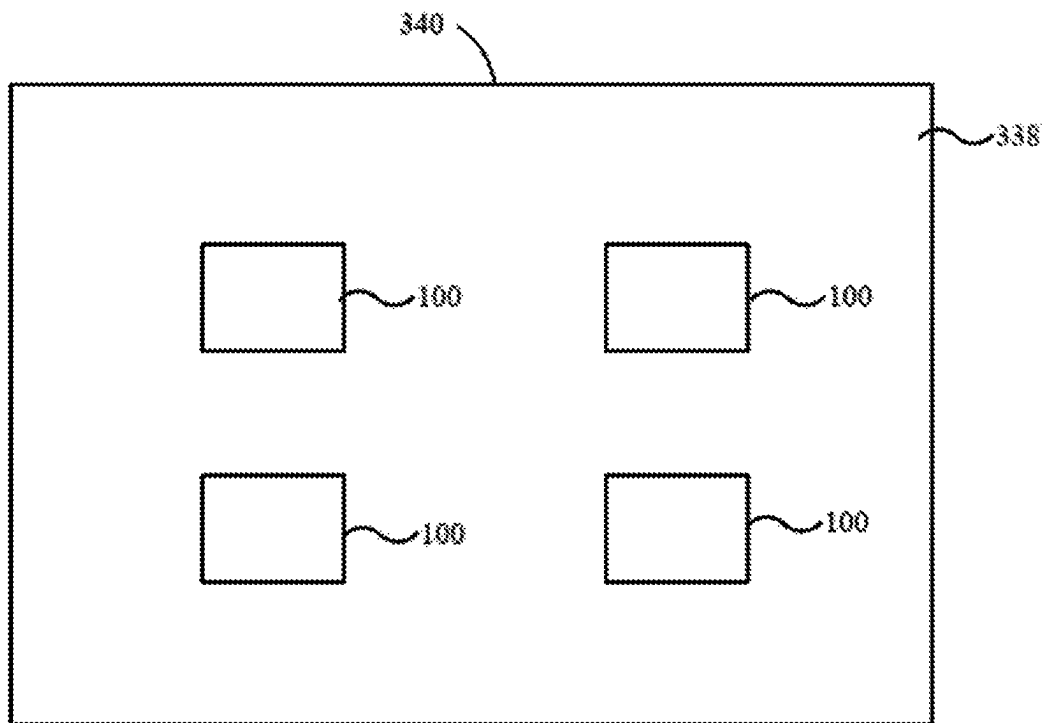


**FIG. 2D**

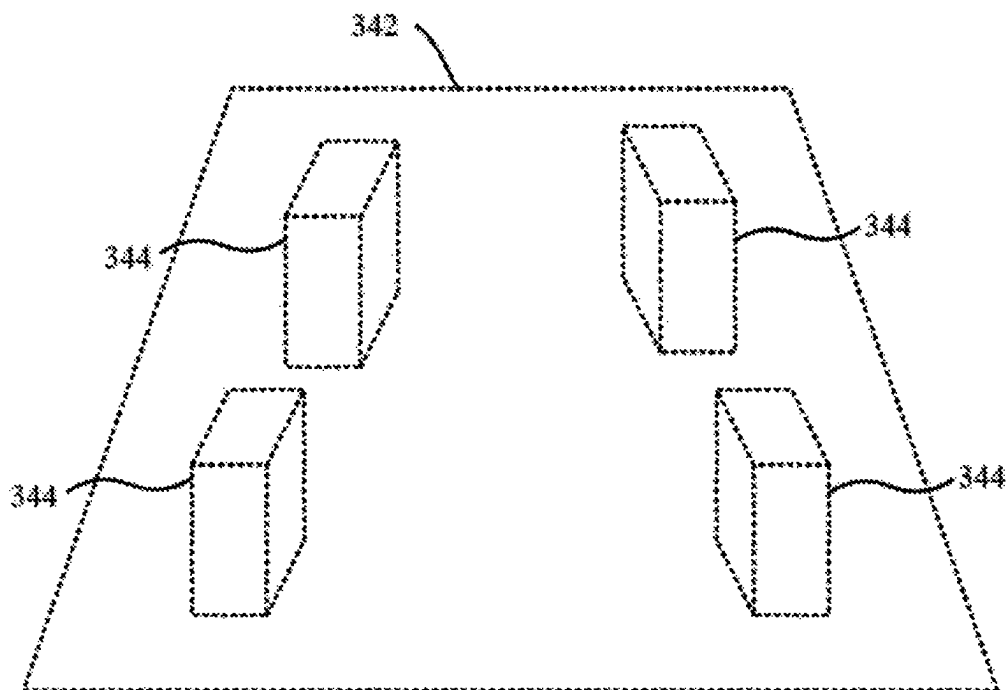


**FIG. 2E**

**FIG. 2F****FIG. 2G**



**FIG. 3A**



**FIG. 3B**



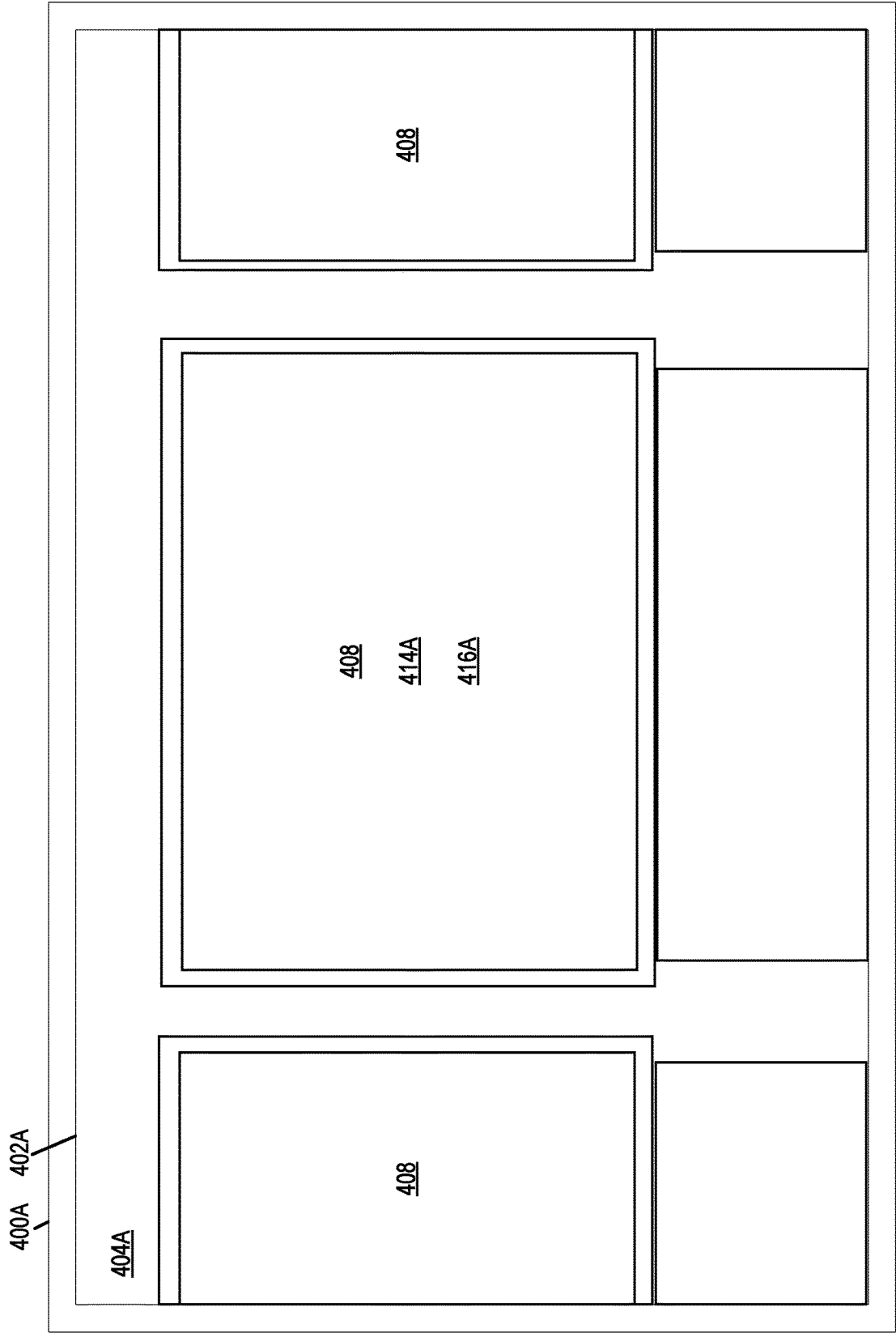
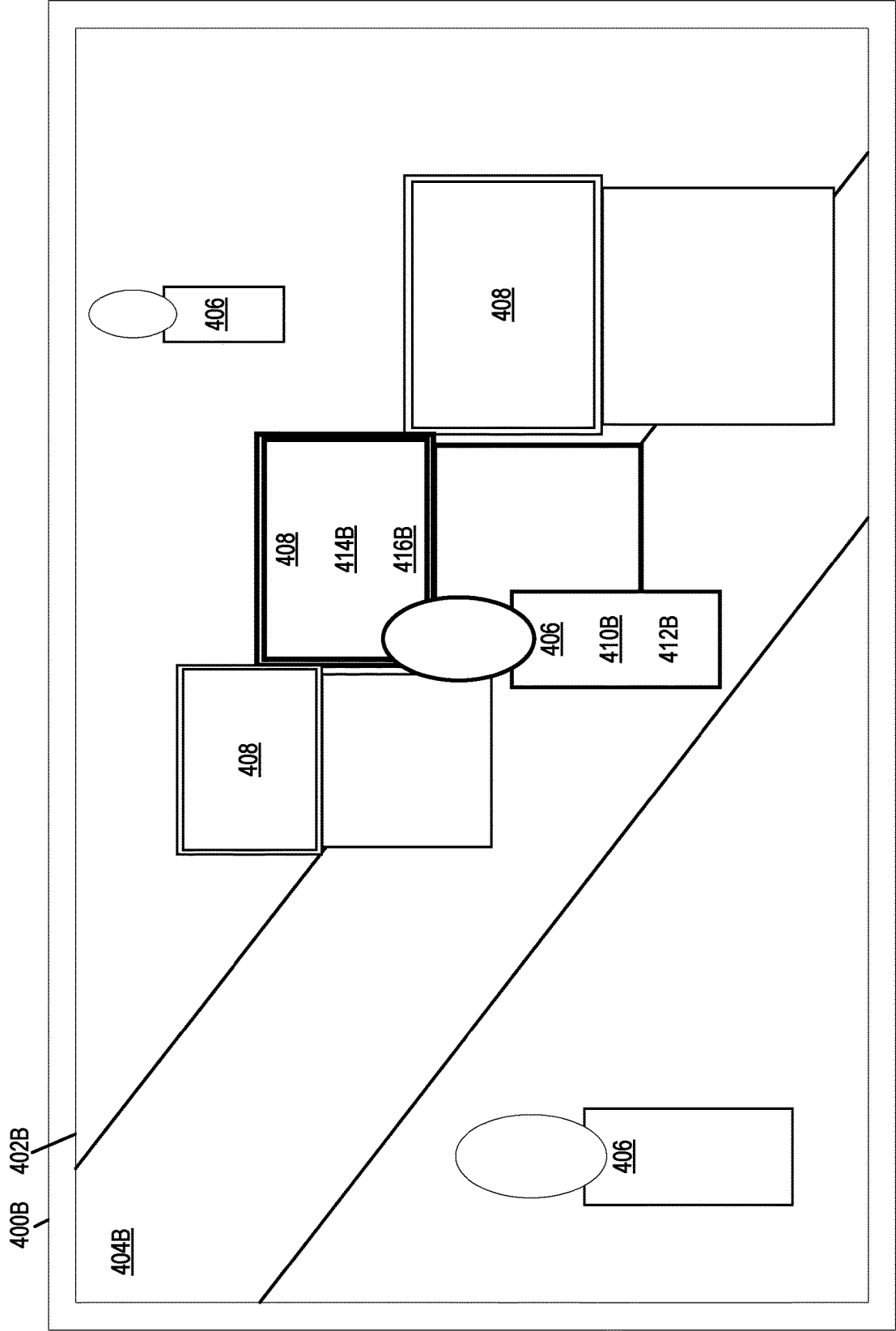


FIG. 4A



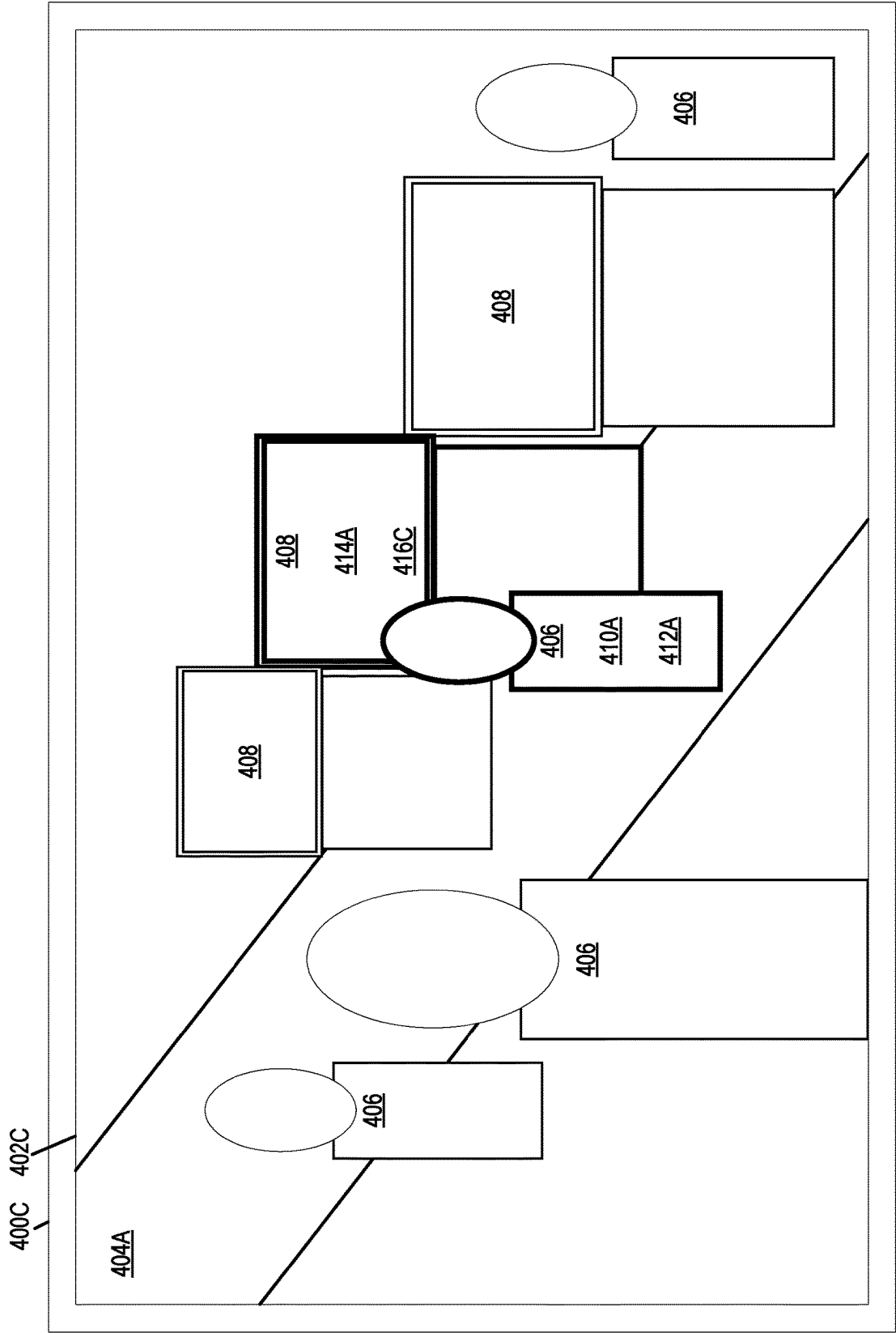


FIG. 4C

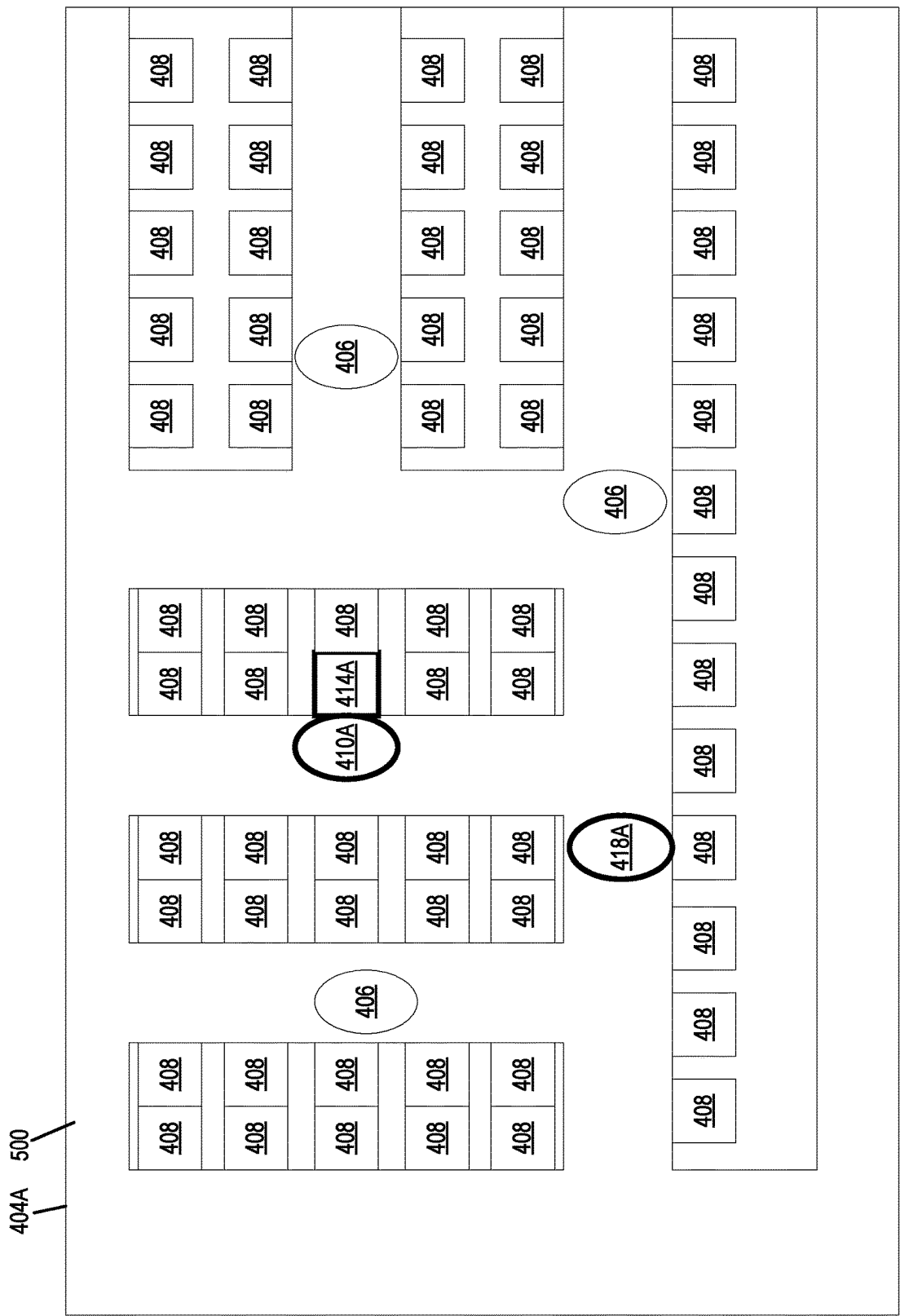


FIG. 5A

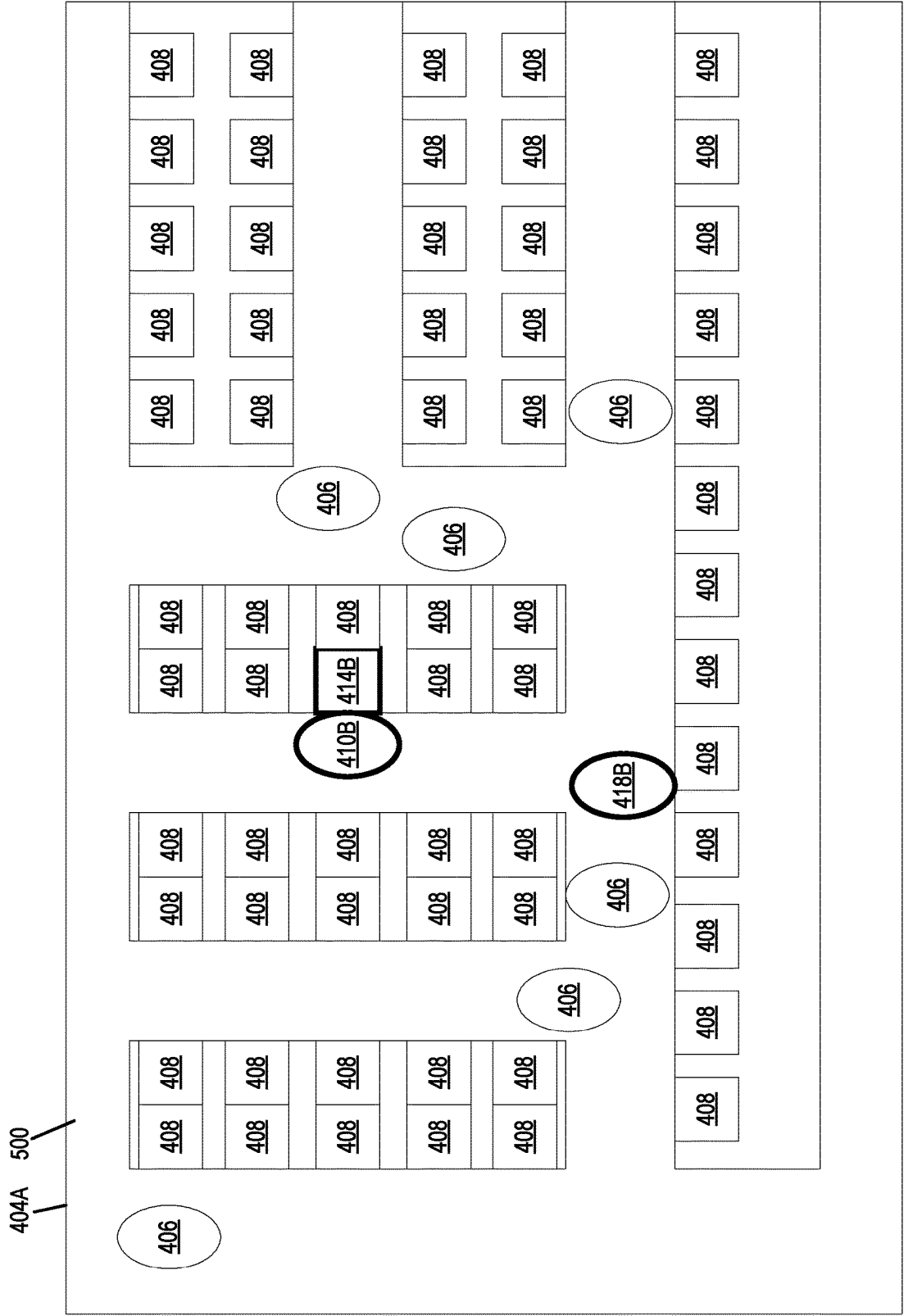
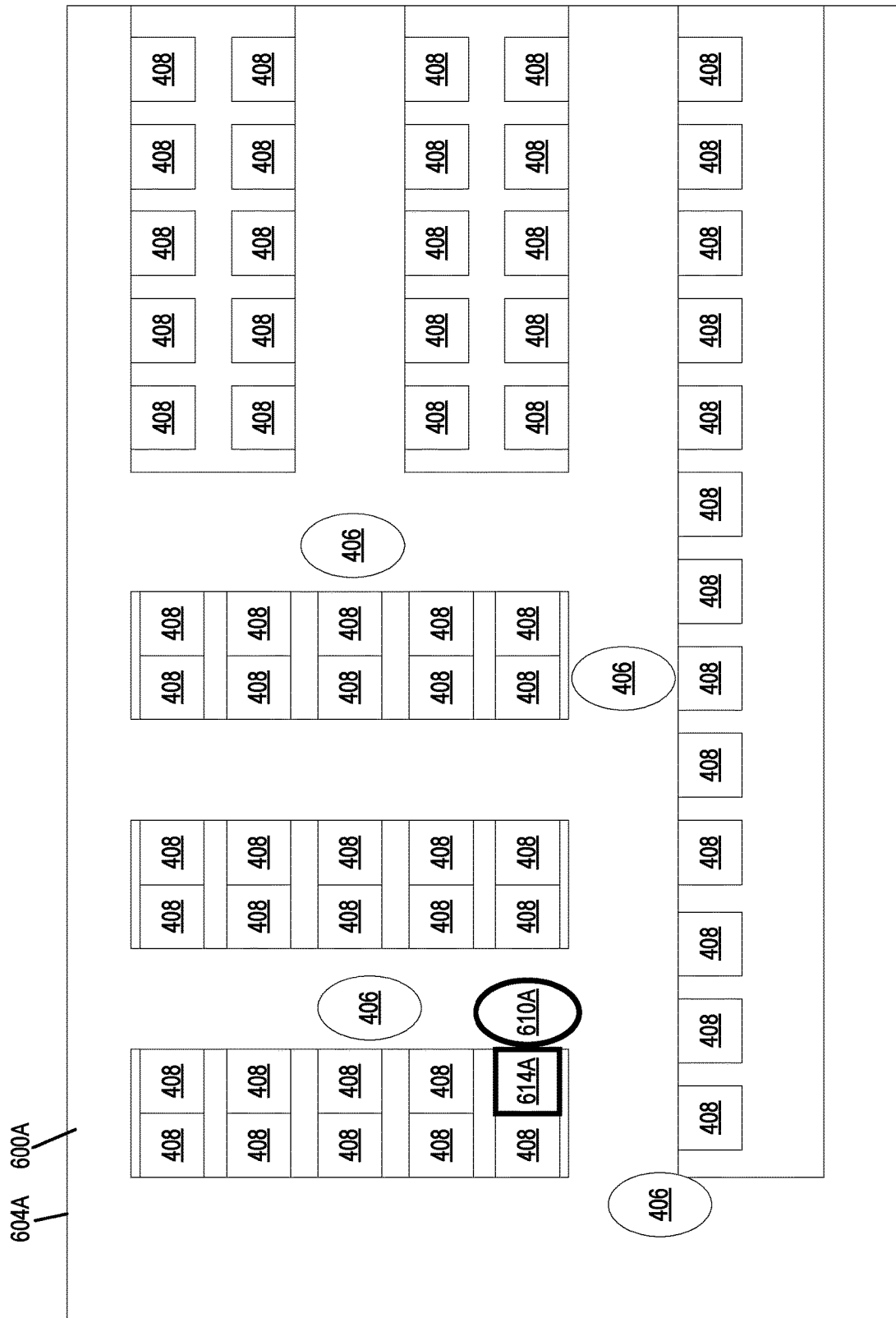


FIG. 5B



**FIG. 6A**

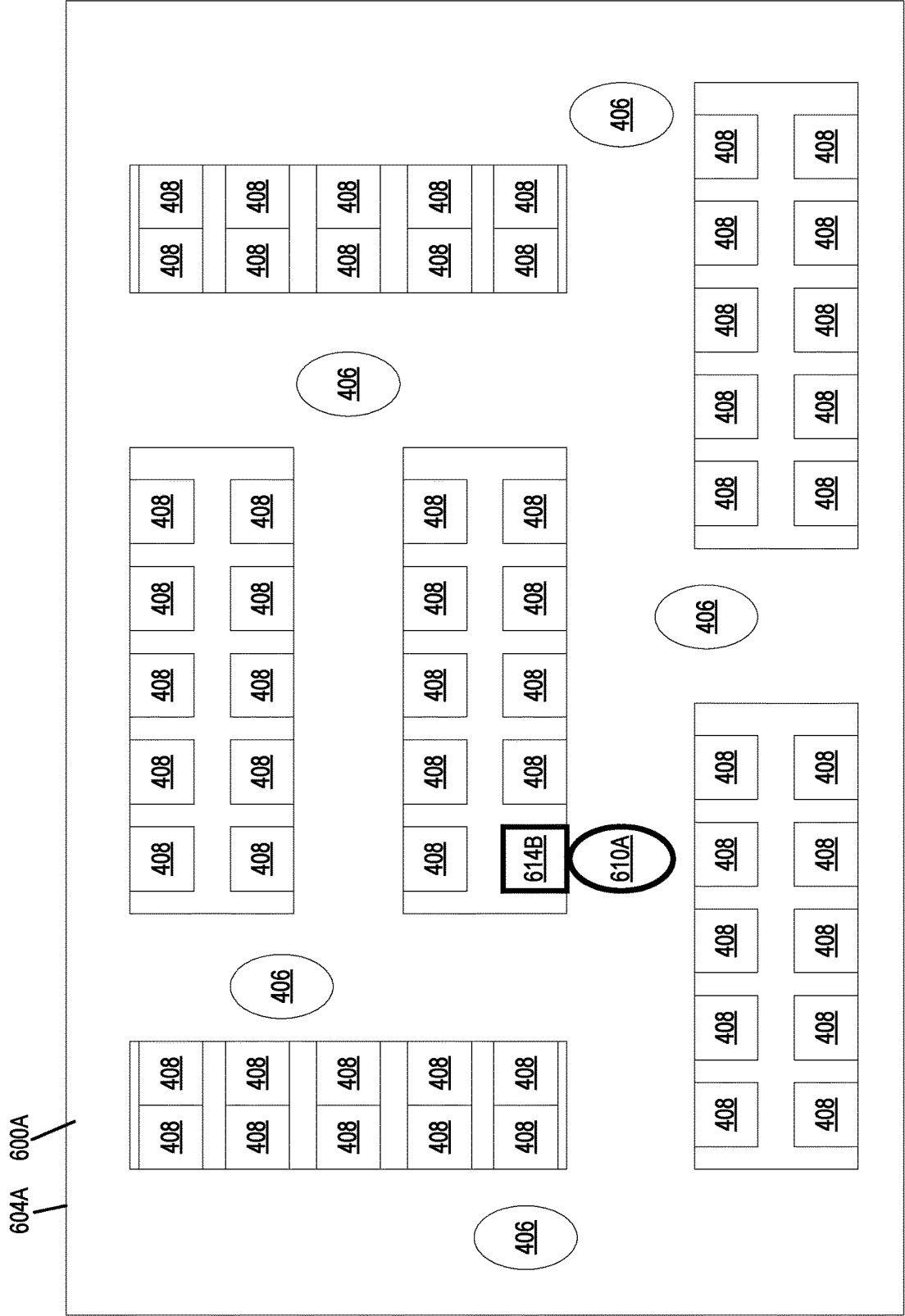
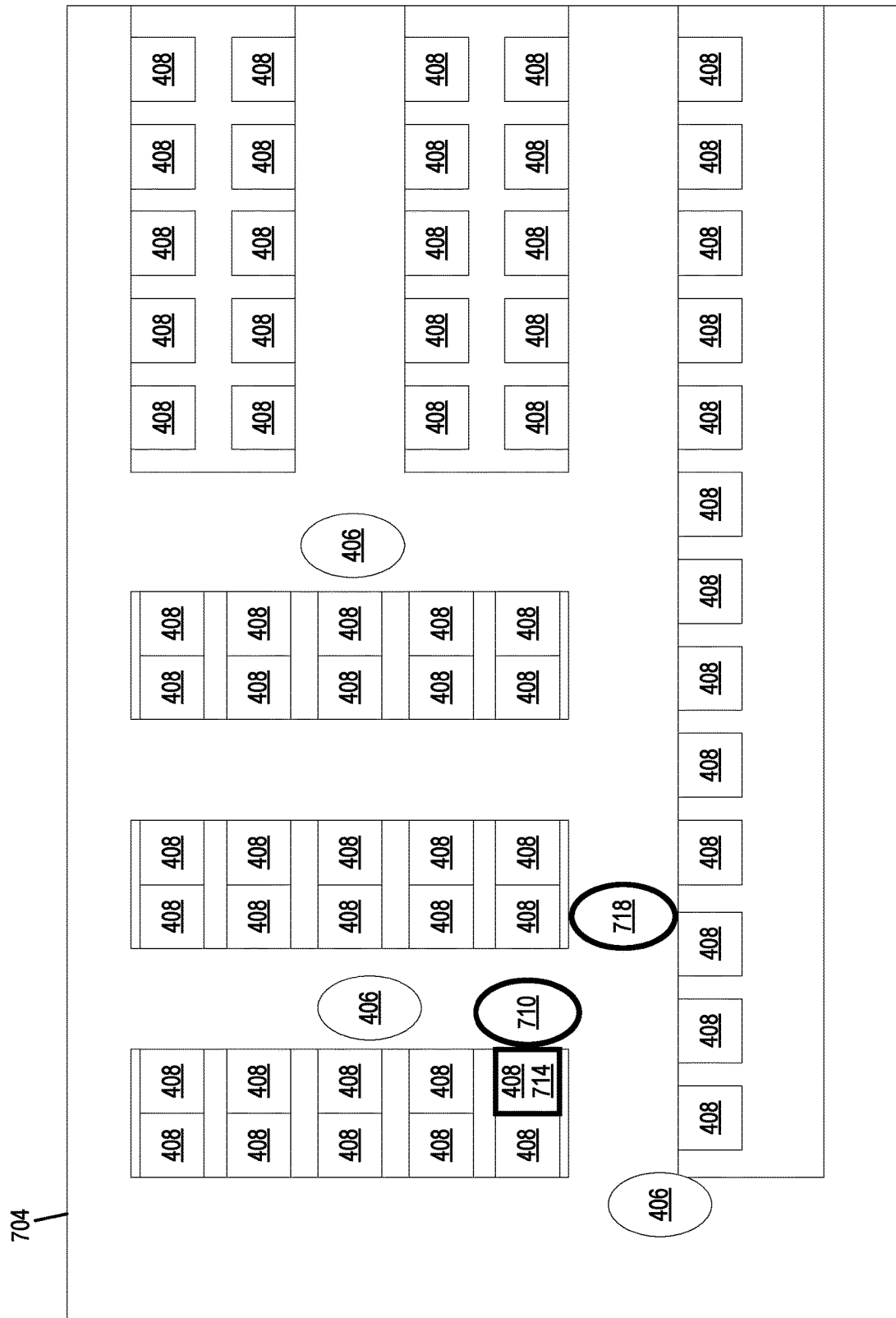
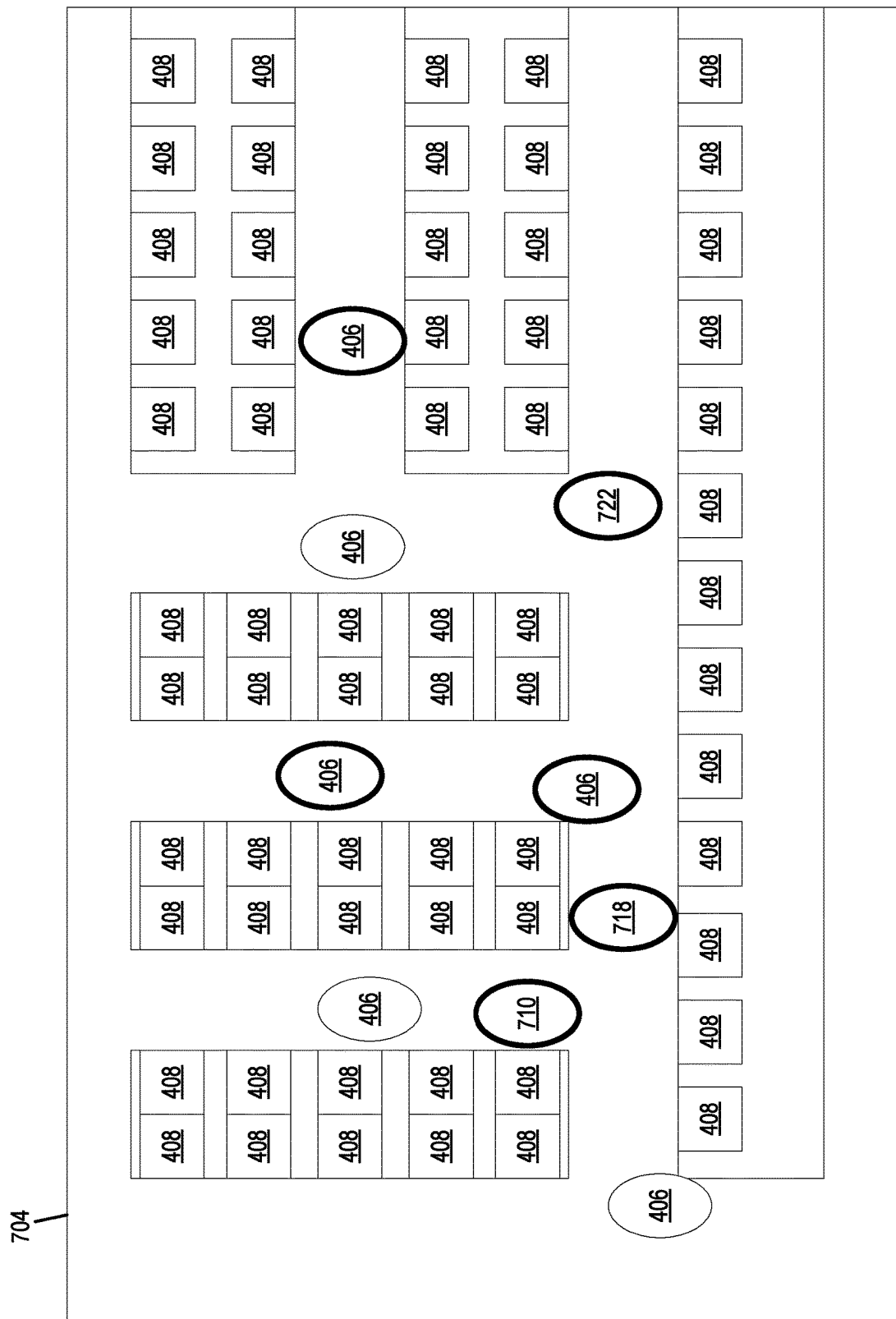


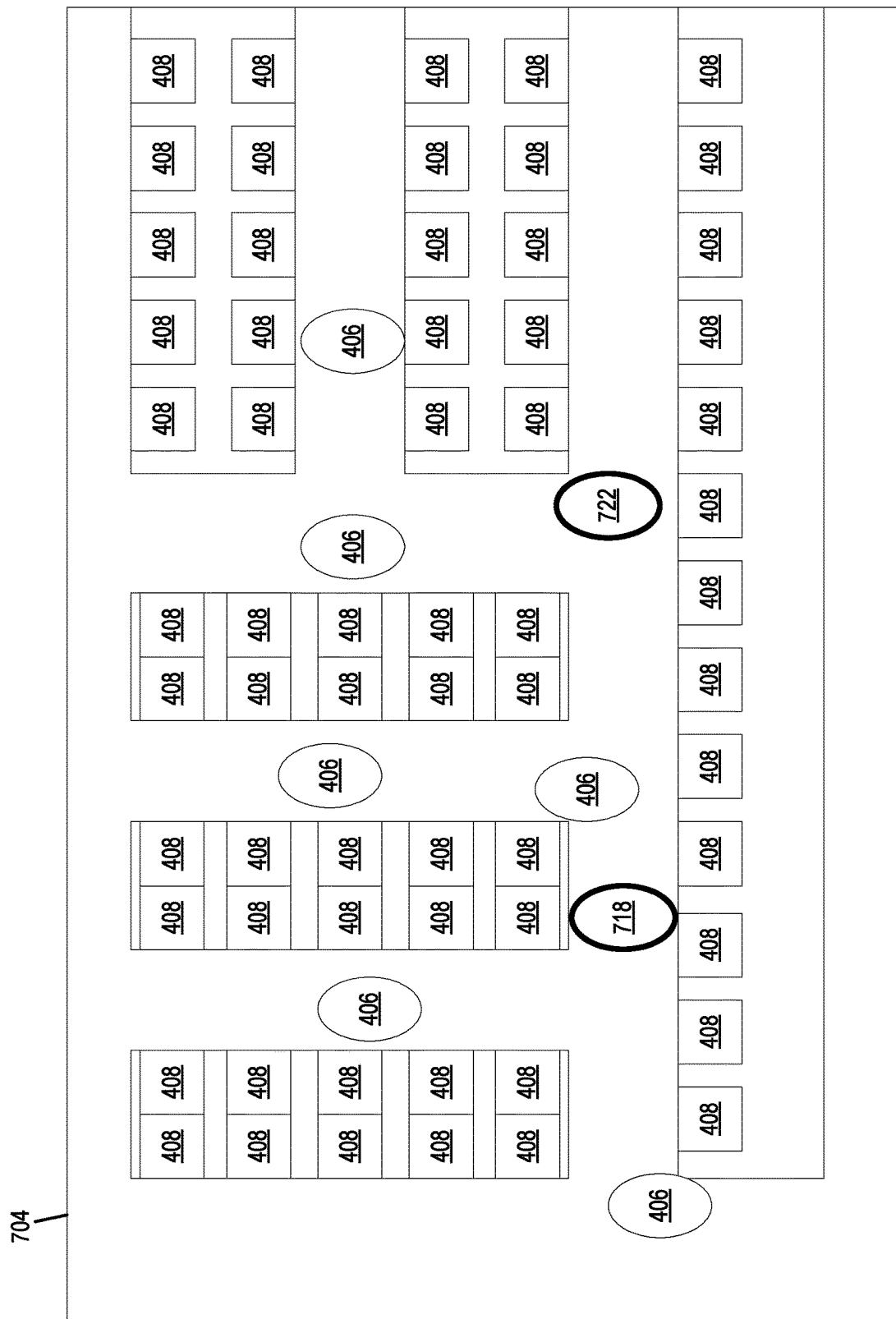
FIG. 6B



**FIG. 7A**







800

Providing, by a processor circuit, a real-time environmental model of a plurality of shared virtual environments (SVEs)

802

Transmitt first display data corresponding to a first SVE of the plurality of SVEs to a first player device worn by a first player, including user display data that causes a display device in the first player device to render a portion of the first SVE based on a virtual orientation of the first player device and a virtual location of the first player in the first SVE

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Transmit second display data corresponding to a second SVE of the plurality of SVEs to a second player device worn by a second player, including user display data that causes a display device in the second player device to render a portion of the second SVE based on a virtual orientation of the second player device and a virtual location of the second player in the second SVE, the portion of the second SVE comprises a virtual person associated with the first player.

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**FIG. 8**

# DISPLAY OF A VIRTUAL PLAYER IN MULTIPLE VIRTUAL REALITY ENVIRONMENTS

## BACKGROUND

Embodiments described herein relate to virtual reality environments, and in particular to display of a virtual player in multiple virtual reality gaming environments, such as virtual casino environments, and related devices, systems, and methods. As the adoption of virtual reality (VR) technology becomes more widespread, many companies are developing persistent virtual worlds that include simulated and/or enhanced reproductions of real world locations, including gaming environments such as casinos for example. There is a need for populate these worlds with virtual persons to provide a more realistic and desirable experience for users. While the amount of digital “real estate” for these worlds is effectively unlimited, conventional virtual worlds may populate these worlds with artificial, e.g., system-controlled or artificial intelligence (AI)-controlled, virtual persons, to provide a more realistic experience for players interacting with these virtual worlds.

## BRIEF SUMMARY

According to some embodiments, a system includes a processor circuit and a memory including machine-readable instructions. When executed by the processor circuit, the instructions cause the processor circuit to provide a real-time environmental model of a plurality of shared virtual environments (SVEs), each SVE comprising a plurality of virtual persons and a plurality of virtual gaming devices. The instructions further cause the processor circuit to transmit first display data corresponding to a first SVE of the plurality of SVEs to a first player device worn by a first player. The first display data includes user display data that causes a display device in the first player device to render a portion of the first SVE based on a virtual orientation of the first player device and a virtual location of the first player in the first SVE. The instructions further cause the processor circuit to transmit second display data corresponding to a second SVE of the plurality of SVEs to a second player device worn by a second player. The second display data includes user display data that causes a display device in the second player device to render a portion of the second SVE based on a virtual orientation of the second player device and a virtual location of the second player in the second SVE. The portion of the second SVE includes a virtual person associated with the first player.

According to some embodiments, a method includes providing, by a processor circuit, a real-time environmental model of a plurality of shared virtual environments (SVEs), each SVE comprising a plurality of virtual persons and a plurality of virtual gaming devices. The method further includes transmitting first display data corresponding to a first SVE of the plurality of SVEs to a first player device worn by a first player. The first display data includes user display data that causes a display device in the first player device to render a portion of the first SVE based on a virtual orientation of the first player device and a virtual location of the first player in the first SVE. The method further includes transmitting second display data corresponding to a second SVE of the plurality of SVEs to a second player device worn by a second player. The second display data includes user display data that causes a display device in the second player device to render a portion of the second SVE based on a

virtual orientation of the second player device and a virtual location of the second player in the second SVE. The portion of the second SVE includes a virtual person associated with the first player.

According to some embodiments, a virtual-reality player device includes a head-wearable frame, a display device coupled to the frame to position the display device in a field of view of a first player, a processor circuit, and a memory including machine-readable instruction. When executed by the processor circuit, the instructions cause the processor circuit to receive first display data corresponding to a first SVE of a plurality of SVEs, each SVE comprising a plurality of virtual persons and a plurality of virtual gaming devices. The instructions further cause the processor circuit to render, by the display device, a portion of the first SVE based on a virtual orientation of the player device and a virtual location of the first player in the first SVE. The instructions further cause the processor circuit to transmit second display data corresponding to a second SVE of the plurality of SVEs to a second player device worn by a second player. The second display data includes user display data that causes a display device in the second player device to render a portion of the second SVE based on a virtual orientation of the second player device and a virtual location of the second player in the second SVE. The portion of the second SVE comprises a virtual person associated with the first player.

## BRIEF DESCRIPTION OF SEVERAL VIEWS OF THE DRAWINGS

FIG. 1 is a schematic block diagram illustrating a network configuration for a plurality of gaming devices according to some embodiments.

FIG. 2A is a perspective view of a gaming device that can be configured according to some embodiments.

FIG. 2B is a schematic block diagram illustrating an electronic configuration for a gaming device according to some embodiments.

FIG. 2C is a schematic block diagram that illustrates various functional modules of a gaming device according to some embodiments.

FIG. 2D is perspective view of a gaming device that can be configured according to some embodiments.

FIG. 2E is a perspective view of a gaming device according to further embodiments.

FIGS. 2F and 2G illustrate VR/AR devices according to various embodiments.

FIG. 3A is a map of a gaming area, such as a casino floor, according to some embodiments.

FIG. 3B is a 3D wireframe model of the gaming area of FIG. 3A, according to some embodiments.

FIGS. 4A-4C are views of a plurality of shared virtual environments (SVEs) corresponding to fields of views (FOVs) of a plurality of players wearing head-mounted virtual reality devices, with virtual players corresponding to the same user appearing in multiple virtual environments simultaneously, according to some embodiments.

FIGS. 5A and 5B are floorplan views of the SVEs of FIGS. 4A-4C having a common floorplan and layout and different subsets of virtual players, according to some embodiments.

FIGS. 6A and 6B are floorplan views of a plurality of SVEs having different floorplans and layouts and different subsets of virtual players, according to some embodiments.

FIGS. 7A-7C are floorplan views of a SVE showing addition and removal of virtual players based on virtual occupancy preferences and rules, according to some embodiments.

FIG. 8 is a flowchart illustrating operations of systems/ methods of facilitating participation in a wagering game between multiple devices, according to some embodiments.

#### DETAILED DESCRIPTION

Embodiments described herein relate to game play features with electronic wagering games, and in particular to display of a virtual player in multiple virtual reality gaming environments, such as virtual casino environments, and related devices, systems, and methods.

Embodiments described herein may be used in connection with virtual environments, e.g., VR environments, as well as real environments with virtual elements, e.g., augmented reality (AR) or mixed reality (MR) environments. For example, one example of a VR implementation may facilitate a player at home using a VR headset to experience the virtual casino or other environment. An example of an AR implementation may have a player in an actual casino environment and using an AR device to augment the experience.

In addition, while many embodiments are described in terms of gaming devices, such as Electronic Gaming Machines (EGMs), aspects of the disclosure may also be used with other types of gaming devices, such as sports wagering terminals, kiosks, table games, etc., and other elements that may be found in a casino floor environment, such as restaurants, retail shopping, or other services, for example.

In a conventional VR implementation, one or more virtual players may be associated with a particular shared virtual environment (SVE), such as a virtual casino environment for example. In this implementation, a virtual person associated with each player would interact with the virtual environment and be visible to other players in the same SVE. Embodiments disclosed herein also contemplate virtual persons associated with a single player existing in multiple SVEs simultaneously, thereby providing the illusion of various virtual casino environments being filled with more real-world players or being busier than they otherwise would be. For example, a player may be playing an EGM in a particular virtual casino, and may also be displayed in a different virtual casino playing the same type of EGM. One advantage of this arrangement is that a greater proportion of virtual persons in these virtual casinos can be associated with real-world players, which may result in greater realism and immersion for other players in the virtual casinos.

In some examples, the behavior of the player in the primary casino environment, i.e., the casino environment with which the player is directly interacting, may be duplicated across additional virtual casinos. The system may also modify behaviors for different environments and/or players interacting with the virtual player in other environments, to better correspond with the environment. For example, a player may be unable to displace another player from a virtual EGM in the other player's primary environment, but may be able to cause the system to displace the other player in the other player's non-primary environment, to free up that particular virtual EGM for the player that is actually interacting with that environment, e.g., by approaching or attempting to play the EGM.

Before discussing these and other embodiments in greater detail, reference will be made to an example of a gaming

system for implementing embodiments disclosed herein. In this regard, FIG. 1 illustrates a gaming system 10 including a plurality of gaming devices 100 is illustrated. As discussed above, the gaming devices 100 may be one type of a variety of different types of gaming devices, such as electronic gaming machines (EGMs), mobile gaming devices, or other devices, for example. The gaming system 10 may be located, for example, on the premises of a gaming establishment, such as a casino. The gaming devices 100, which are typically situated on a casino floor, may be in communication with each other and/or at least one central controller 40 through a data communication network 50 that may include a remote communication link. The data communication network 50 may be a private data communication network that is operated, for example, by the gaming facility that operates the gaming devices 100. Communications over the data communication network 50 may be encrypted for security. The central controller 40 may be any suitable server or computing device which includes at least one processing circuit and at least one memory or storage device. Each gaming device 100 may include a processing circuit that transmits and receives events, messages, commands or any other suitable data or signal between the gaming device 100 and the central controller 40. The gaming device processing circuit is operable to execute such communicated events, messages or commands in conjunction with the operation of the gaming device 100. Moreover, the processing circuit of the central controller 40 is configured to transmit and receive events, messages, commands or any other suitable data or signal between the central controller 40 and each of the individual gaming devices 100. In some embodiments, one or more of the functions of the central controller 40 may be performed by one or more gaming device processing circuits. Moreover, in some embodiments, one or more of the functions of one or more gaming device processing circuits as disclosed herein may be performed by the central controller 40.

A wireless access point 60 provides wireless access to the data communication network 50. The wireless access point 60 may be connected to the data communication network 50 as illustrated in FIG. 1, and/or may be connected directly to the central controller 40 or another server connected to the data communication network 50.

A player tracking server 45 may also be connected through the data communication network 50. The player tracking server 45 may manage a player tracking account that tracks the player's gameplay and spending and/or other player preferences and customizations, manages loyalty awards for the player, manages funds deposited or advanced on behalf of the player, and other functions. Player information managed by the player tracking server 45 may be stored in a player information database 47.

As further illustrated in FIG. 1, the gaming system 10 may include a ticket server 90 that is configured to print and/or dispense wagering tickets. The ticket server 90 may be in communication with the central controller 40 through the data communication network 50. Each ticket server 90 may include a processing circuit that transmits and receives events, messages, commands or any other suitable data or signal between the ticket server 90 and the central controller 40. The ticket server 90 processing circuit may be operable to execute such communicated events, messages or commands in conjunction with the operation of the ticket server 90. Moreover, in some embodiments, one or more of the functions of one or more ticket server 90 processing circuits as disclosed herein may be performed by the central controller 40.

The gaming devices **100** communicate with one or more elements of the gaming system **10** to coordinate providing wagering games and other functionality. For example, in some embodiments, the gaming device **100** may communicate directly with the ticket server **90** over a wireless interface **62**, which may be a WiFi link, a Bluetooth link, a near field communications (NFC) link, etc. In other embodiments, the gaming device **100** may communicate with the data communication network **50** (and devices connected thereto, including other gaming devices **100**) over a wireless interface **64** with the wireless access point **60**. The wireless interface **64** may include a WiFi link, a Bluetooth link, an NFC link, etc. In still further embodiments, the gaming devices **100** may communicate simultaneously with both the ticket server **90** over the wireless interface **66** and the wireless access point **60** over the wireless interface **64**. Some embodiments provide that gaming devices **100** may communicate with other gaming devices over a wireless interface **64**. In these embodiments, wireless interface **62**, wireless interface **64** and wireless interface **66** may use different communication protocols and/or different communication resources, such as different frequencies, time slots, spreading codes, etc.

The wireless interfaces **62**, **66** allow a plurality of virtual reality (VR) and/or augmented reality (AR) devices, referred to herein as VR/AR devices **200**, to coordinate the generation and rendering of VR and/or AR images to the player. As used herein, VR/AR devices **200** may include VR and/or AR functionality, as desired. In some embodiments, the gaming system **10** includes a VR/AR controller **114**. The VR/AR controller **114** may be a computing system that communicates through the data communication network **50** with the EGMs **100** and the VR devices **200** to coordinate the generation and rendering of virtual images to one or more players using the VR devices **200**. The VR/AR controller **114** may be implemented within or separately from the central controller **40**.

In some embodiments, the VR/AR controller **114** may coordinate the generation and display of the virtual images of the same virtual object to more than one player by more than one VR/AR device **200**. As described in more detail below, this may enable multiple players to interact with the same virtual object together in real time. This feature can be used to provide a shared multiplayer experience to multiple players at the same time.

Moreover, in some embodiments, the VR/AR controller **114** may coordinate the generation and display of the same virtual object to players at different physical locations, as will be described in more detail below.

The VR/AR controller **114** may store a three-dimensional wireframe map of a gaming area, such as a casino floor, and may provide the three-dimensional wireframe map to the VR/AR devices **200**. The wireframe map may store various information about EGMs in the gaming area, such as the identity, type and location of various types of EGMs. The three-dimensional wireframe map may enable a VR/AR device **200** to more quickly and accurately determine its position and/or orientation within the gaming area, and also may enable the VR/AR device **200** to assist the player in navigating the gaming area while using the VR/AR device **200**. The generation of three-dimensional wireframe maps is described in more detail below.

In some embodiments, at least some processing of virtual images and/or objects that are rendered by the VR devices **200** may be performed by the VR/AR controller **114**, thereby offloading at least some processing requirements from the VR devices **200**.

Embodiments herein may include different types of gaming devices. One example of a gaming device includes a gaming device **100** that can use gesture, voice, and/or touch-based inputs according to various embodiments is illustrated in FIGS. **2A**, **2B**, and **2C** in which FIG. **2A** is a perspective view of a gaming device **100** illustrating various physical features of the device, FIG. **2B** is a functional block diagram that schematically illustrates an electronic relationship of various elements of the gaming device **100**, and FIG. **2C** illustrates various functional modules that can be stored in a memory device of the gaming device **100**. The embodiments shown in FIGS. **2A** to **2C** are provided as examples for illustrative purposes only. It will be appreciated that gaming devices may come in many different shapes, sizes, layouts, form factors, and configurations, and with varying numbers and types of input and output devices, and that embodiments are not limited to the particular gaming device structures described herein.

Gaming devices **100** typically include a number of standard features, many of which are illustrated in FIGS. **2A** and **2B**. For example, referring to FIG. **2A**, a gaming device **100** (which is an EGM **160** in this embodiment) may include a support structure, housing **105** (e.g., cabinet) which provides support for a plurality of displays, inputs, outputs, controls and other features that enable a player to interact with the gaming device **100**.

The gaming device **100** illustrated in FIG. **2A** includes a number of display devices, including a primary display device **116** located in a central portion of the housing **105** and a secondary display device **118** located in an upper portion of the housing **105**. A plurality of game components **155** are displayed on a display screen **117** of the primary display device **116**. It will be appreciated that one or more of the display devices **116**, **118** may be omitted, or that the display devices **116**, **118** may be combined into a single display device. The gaming device **100** may further include a player tracking display **142**, a credit display **120**, and a bet display **122**. The credit display **120** displays a player's current number of credits, cash, account balance or the equivalent. The bet display **122** displays a player's amount wagered. Locations of these displays are merely illustrative as any of these displays may be located anywhere on the gaming device **100**.

The player tracking display **142** may be used to display a service window that allows the player to interact with, for example, their player loyalty account to obtain features, bonuses, comps, etc. In other embodiments, additional display screens may be provided beyond those illustrated in FIG. **2A**. In some embodiments, one or more of the player tracking display **142**, the credit display **120** and the bet display **122** may be displayed in one or more portions of one or more other displays that display other game related visual content. For example, one or more of the player tracking display **142**, the credit display **120** and the bet display **122** may be displayed in a picture in a picture on one or more displays.

The gaming device **100** may further include a number of input devices **130** that allow a player to provide various inputs to the gaming device **100**, either before, during or after a game has been played. The gaming device may further include a game play initiation button **132** and a cashout button **134**. The cashout button **134** is utilized to receive a cash payment or any other suitable form of payment corresponding to a quantity of remaining credits of a credit display.

In some embodiments, one or more input devices of the gaming device **100** are one or more game play activation

devices that are each used to initiate a play of a game on the gaming device **100** or a sequence of events associated with the gaming device **100** following appropriate funding of the gaming device **100**. The example gaming device **100** illustrated in FIGS. 2A and 2B includes a game play activation device in the form of a game play initiation button **132**. It should be appreciated that, in other embodiments, the gaming device **100** begins game play automatically upon appropriate funding rather than upon utilization of the game play activation device.

In some embodiments, one or more input device **130** of the gaming device **100** may include wagering or betting functionality. For example, a maximum wagering or betting function may be provided that, when utilized, causes a maximum wager to be placed. Another such wagering or betting function is a repeat the bet device that, when utilized, causes the previously-placed wager to be placed. A further such wagering or betting function is a bet one function. A bet is placed upon utilization of the bet one function. The bet is increased by one credit each time the bet one device is utilized. Upon the utilization of the bet one function, a quantity of credits shown in a credit display (as described below) decreases by one, and a number of credits shown in a bet display (as described below) increases by one.

In some embodiments, as shown in FIG. 2B, the input device(s) **130** may include and/or interact with additional components, such as gesture sensors **156** for gesture input devices, and/or a touch-sensitive display that includes a digitizer **152** and a touchscreen controller **154** for touch input devices, as disclosed herein. The player may interact with the gaming device **100** by touching virtual buttons on one or more of the display devices **116**, **118**, **140**. Accordingly, any of the above-described input devices, such as the input device **130**, the game play initiation button **132** and/or the cashout button **134** may be provided as virtual buttons or regions on one or more of the display devices **116**, **118**, **140**.

Referring briefly to FIG. 2B, operation of the primary display device **116**, the secondary display device **118** and the player tracking display **142** may be controlled by a video controller **30** that receives video data from a processing circuit **12** or directly from a memory device **14** and displays the video data on the display screen. The credit display **120** and the bet display **122** are typically implemented as simple liquid crystal display (LCD) or light emitting diode (LED) displays that display a number of credits available for wagering and a number of credits being wagered on a particular game. Accordingly, the credit display **120** and the bet display **122** may be driven directly by the processing circuit **12**. In some embodiments however, the credit display **120** and/or the bet display **122** may be driven by the video controller **30**.

Referring again to FIG. 2A, the display devices **116**, **118**, **140** may include, without limitation: a cathode ray tube, a plasma display, an LCD, a display based on LEDs, a display based on a plurality of organic light-emitting diodes (OLEDs), a display based on polymer light-emitting diodes (PLEDs), a display based on a plurality of surface-conduction electron-emitters (SEDs), a display including a projected and/or reflected image, or any other suitable electronic device or display mechanism. In certain embodiments, as described above, the display devices **116**, **118**, **140** may include a touch-screen with an associated touchscreen controller **154** and digitizer **152**. The display devices **116**, **118**, **140** may be of any suitable size, shape, and/or configuration. The display devices **116**, **118**, **140** may include flat or curved display surfaces.

The display devices **116**, **118**, **140** and video controller **30** of the gaming device **100** are generally configured to display one or more game and/or non-game images, symbols, and indicia. In certain embodiments, the display devices **116**, **118**, **140** of the gaming device **100** are configured to display any suitable visual representation or exhibition of the movement of objects; dynamic lighting; video images; images of people, characters, places, things, and faces of cards; and the like. In certain embodiments, the display devices **116**, **118**, **140** of the gaming device **100** are configured to display one or more virtual reels, one or more virtual wheels, and/or one or more virtual dice. In other embodiments, certain of the displayed images, symbols, and indicia are in mechanical form. That is, in these embodiments, the display device **116**, **118**, **140** includes any electromechanical device, such as one or more rotatable wheels, one or more reels, and/or one or more dice, configured to display at least one or a plurality of game or other suitable images, symbols, or indicia.

The gaming device **100** also includes various features that enable a player to deposit credits in the gaming device **100** and withdraw credits from the gaming device **100**, such as in the form of a payout of winnings, credits, etc. For example, the gaming device **100** may include a bill/ticket dispenser **136**, a bill/ticket acceptor **128**, and a coin acceptor **126** that allows the player to deposit coins into the gaming device **100**.

As illustrated in FIG. 2A, the gaming device **100** may also include a currency dispenser **137** that may include a note dispenser configured to dispense paper currency and/or a coin generator configured to dispense coins or tokens in a coin payout tray.

The gaming device **100** may further include one or more speakers **150** controlled by one or more sound cards **28** (FIG. 2B). The gaming device **100** illustrated in FIG. 2A includes a pair of speakers **150**. In other embodiments, additional speakers, such as surround sound speakers, may be provided within or on the housing **105**. Moreover, the gaming device **100** may include built-in seating with integrated headrest speakers.

In various embodiments, the gaming device **100** may generate dynamic sounds coupled with attractive multimedia images displayed on one or more of the display devices **116**, **118**, **140** to provide an audio-visual representation or to otherwise display full-motion video with sound to attract players to the gaming device **100** and/or to engage the player during gameplay. In certain embodiments, the gaming device **100** may display a sequence of audio and/or visual attraction messages during idle periods to attract potential players to the gaming device **100**. The videos may be customized to provide any appropriate information.

The gaming device **100** may further include a card reader **138** that is configured to read magnetic stripe cards, such as player loyalty/tracking cards, chip cards, and the like. In some embodiments, a player may insert an identification card into a card reader of the gaming device. In some embodiments, the identification card is a smart card having a programmed microchip or a magnetic strip coded with a player's identification, credit totals (or related data) and other relevant information. In other embodiments, a player may carry a portable device, such as a cell phone, a radio frequency identification tag or any other suitable wireless device, which communicates a player's identification, credit totals (or related data) and other relevant information to the gaming device. In some embodiments, money may be transferred to a gaming device through electronic funds transfer. When a player funds the gaming device, the processing circuit determines the amount of funds entered and

displays the corresponding amount on the credit or other suitable display as described above.

In some embodiments, the gaming device 100 may include an electronic payout device or module configured to fund an electronically recordable identification card or smart card or a bank or other account via an electronic funds transfer to or from the gaming device 100.

FIG. 2B is a block diagram that illustrates logical and functional relationships between various components of a gaming device 100. It should also be understood that components described in FIG. 2B may also be used in other computing devices, as desired, such as mobile computing devices for example. As shown in FIG. 2B, the gaming device 100 may include a processing circuit 12 that controls operations of the gaming device 100. Although illustrated as a single processing circuit, multiple special purpose and/or general purpose processors and/or processor cores may be provided in the gaming device 100. For example, the gaming device 100 may include one or more of a video processor, a signal processor, a sound processor and/or a communication controller that performs one or more control functions within the gaming device 100. The processing circuit 12 may be variously referred to as a “controller,” “microcontroller,” “microprocessor” or simply a “computer.” The processor may further include one or more application-specific integrated circuits (ASICs).

Various components of the gaming device 100 are illustrated in FIG. 2B as being connected to the processing circuit 12. It will be appreciated that the components may be connected to the processing circuit 12 through a system bus 151, a communication bus and controller, such as a universal serial bus (USB) controller and USB bus, a network interface, or any other suitable type of connection.

The gaming device 100 further includes a memory device 14 that stores one or more functional modules 20. Various functional modules 20 of the gaming device 100 will be described in more detail below in connection with FIG. 2D.

The memory device 14 may store program code and instructions, executable by the processing circuit 12, to control the gaming device 100. The memory device 14 may also store other data such as image data, event data, player input data, random or pseudorandom number generators, pay-table data or information and applicable game rules that relate to the play of the gaming device. The memory device 14 may include random access memory (RAM), which can include non-volatile RAM (NVRAM), magnetic RAM (ARAM), ferroelectric RAM (FeRAM) and other forms as commonly understood in the gaming industry. In some embodiments, the memory device 14 may include read only memory (ROM). In some embodiments, the memory device 14 may include flash memory and/or EEPROM (electrically erasable programmable read only memory). Any other suitable magnetic, optical and/or semiconductor memory may operate in conjunction with the gaming device disclosed herein.

The gaming device 100 may further include a data storage 22, such as a hard disk drive or flash memory. The data storage 22 may store program data, player data, audit trail data or any other type of data. The data storage 22 may include a detachable or removable memory device, including, but not limited to, a suitable cartridge, disk, CD ROM, Digital Video Disc (“DVD”) or USB memory device.

The gaming device 100 may include a communication adapter 26 that enables the gaming device 100 to communicate with remote devices over a wired and/or wireless communication network, such as a local area network (LAN), wide area network (WAN), cellular communication

network, or other data communication network. The communication adapter 26 may further include circuitry for supporting short range wireless communication protocols, such as Bluetooth and/or NFC that enable the gaming device 100 to communicate, for example, with a mobile communication device operated by a player.

The gaming device 100 may include one or more internal or external communication ports that enable the processing circuit 12 to communicate with and to operate with internal or external peripheral devices, such as eye tracking devices, position tracking devices, cameras, accelerometers, arcade sticks, bar code readers, bill validators, biometric input devices, bonus devices, button panels, card readers, coin dispensers, coin hoppers, display screens or other displays or video sources, expansion buses, information panels, keypads, lights, mass storage devices, microphones, motion sensors, motors, printers, reels, Small Computer System Interface (“SCSI”) ports, solenoids, speakers, thumb drives, ticket readers, touch screens, trackballs, touchpads, wheels, and wireless communication devices. In some embodiments, internal or external peripheral devices may communicate with the processing circuit through a USB hub (not shown) connected to the processing circuit 12.

In some embodiments, the gaming device 100 may include a sensor, such as a camera 127, in communication with the processing circuit 12 (and possibly controlled by the processing circuit 12) that is selectively positioned to acquire an image of a player actively using the gaming device 100 and/or the surrounding area of the gaming device 100. In one embodiment, the camera 127 may be configured to selectively acquire still or moving (e.g., video) images and may be configured to acquire the images in either an analog, digital or other suitable format. The display devices 116, 118, 140 may be configured to display the image acquired by the camera 127 as well as display the visible manifestation of the game in split screen or picture-in-picture fashion. For example, the camera 127 may acquire an image of the player and the processing circuit 12 may incorporate that image into the primary and/or secondary game as a game image, symbol or indicia.

Various functional modules of that may be stored in a memory device 14 of a gaming device 100 are illustrated in FIG. 2C. Referring to FIG. 2C, the gaming device 100 may include in the memory device 14 a game module 20A that includes program instructions and/or data for operating a hybrid wagering game as described herein. The gaming device 100 may further include a player tracking module 20B, an electronic funds transfer module 20C, an input device interface 20D, an audit/reporting module 20E, a communication module 20F, an operating system kernel 20G and a random number generator 20H. The player tracking module 20B keeps track of the play of a player. The electronic funds transfer module 20C communicates with a back end server or financial institution to transfer funds to and from an account associated with the player. The input device interface 20D interacts with input devices, such as the input device 130, as described in more detail below. The communication module 20F enables the gaming device 100 to communicate with remote servers and other gaming devices using various secure communication interfaces. The operating system kernel 20G controls the overall operation of the gaming device 100, including the loading and operation of other modules. The random number generator 20H generates random or pseudorandom numbers for use in the operation of the hybrid games described herein.

In some embodiments, a gaming device 100 includes a personal device, such as a desktop computer, a laptop



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computer, a mobile device, a tablet computer or computing device, a personal digital assistant (PDA), or other portable computing devices. In some embodiments, the gaming device **100** may be operable over a wireless network, such as part of a wireless gaming system. In such embodiments, the gaming machine may be a hand-held device, a mobile device or any other suitable wireless device that enables a player to play any suitable game at a variety of different locations. It should be appreciated that a gaming device or gaming machine as disclosed herein may be a device that has obtained approval from a regulatory gaming commission or a device that has not obtained approval from a regulatory gaming commission.

For example, referring to FIG. 2D, a gaming device **100** (which is a mobile gaming device **170** in this embodiment) may be implemented as a handheld device including a compact housing **105** on which is mounted a touchscreen display device **116** including a digitizer **152**. As described in greater detail with respect to FIG. 3 below, one or more input devices **130** may be included for providing functionality of for embodiments described herein. A camera **127** may be provided in a front face of the housing **105**. The housing **105** may include one or more speakers **150**. In the gaming device **100**, various input buttons described above, such as the cashout button, gameplay activation button, etc., may be implemented as soft buttons on the touchscreen display device **116** and/or input device **130**. In this embodiment, the input device **130** is integrated into the touchscreen display device **116**, but it should be understood that the input device may also, or alternatively, be separate from the display device **116**. Moreover, the gaming device **100** may omit certain features, such as a bill acceptor, a ticket generator, a coin acceptor or dispenser, a card reader, secondary displays, a bet display, a credit display, etc. Credits can be deposited in or transferred from the gaming device **100** electronically.

FIG. 2E illustrates a standalone gaming device **100** (which is an EGM **160** in this embodiment) having a different form factor from the EGM **160** illustrated in FIG. 2A. In particular, the gaming device **100** is characterized by having a large, high aspect ratio, curved primary display device **116** provided in the housing **105**, with no secondary display device. The primary display device **116** may include a digitizer **152** to allow touchscreen interaction with the primary display device **116**. The gaming device **100** may further include a player tracking display **142**, an input device **130**, a bill/ticket acceptor **128**, a card reader **138**, and a bill/ticket dispenser **136**. The gaming device **100** may further include one or more cameras **127** to enable facial recognition and/or motion tracking.

FIG. 2F illustrates a virtual reality (VR) viewer **200A** implemented as a 3D headset including a pair of displays **218** on which images of virtual objects may be displayed. The viewer **200A** includes a head-wearable frame **202**, with the displays **218** coupled to the frame **202** to position the display device in a field of view of user wearing the viewer **200A**. Different stereoscopic images may be displayed on the displays **218** to create an appearance of depth. The VR viewer **200A** may include a plurality of sensors **220** that the device uses to determine a position, orientation, and/or movement of the viewer **200A**, which may be used to determine a position, orientation, and/or direction of movement within an SVE.

The viewer **200A** may further include other sensors, such as a gyroscopic sensor, a GPS sensor, one or more accelerometers, and/or other sensors that allow the viewer **200A** to determine its position and orientation in space. In some embodiments, the viewer **200A** may include one or more

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cameras that allow the viewer **200A** to determine its position and/or orientation in space using visual simultaneous localization and mapping (VSLAM). viewer **200A** may further include one or more microphones and/or speakers that allow the user to interact audially with the device.

In some embodiments, a viewer may also include semi-transparent lenses that allow the user to see both the real world as well as the 3D image rendered on the lenses, e.g., to provide an augmented reality (AR) experience. The viewer may also include additional cameras or other sensors to obtain a live video signal for building a 3D model of the space around the user. The viewer may also generate a 3D image of an object to display to the user that takes into account the real world objects around the user and allows the user to interact with the 3D object.

Referring to FIG. 2G, an augmented reality (AR) viewer **200B** may be implemented as a pair of glasses including a transparent prismatic display **222** that displays an image to a single eye of the user. Such a device may be capable of displaying images to the user while allowing the user to see the world around the user, and as such can be used as an AR device.

In other embodiments, a VR and/or AR viewer may be implemented using a virtual retinal display device that raster scans an image directly onto the retina of the user. In still further embodiments, a VR and/or AR viewer may be implemented using a mobile wireless device, such as the mobile gaming device **170** of FIG. 2D above, a mobile telephone, a tablet computing device, and/or a personal digital assistant, etc.

Although illustrated as certain gaming devices, such as electronic gaming machines (EGMs), mobile gaming devices, VR/AR headsets, etc., functions and/or operations as described herein may also include wagering stations that may include electronic game tables, conventional game tables including those involving cards, dice and/or roulette, and/or other wagering stations such as sports book stations, video poker games, skill-based games, virtual casino-style table games, or other casino or non-casino style games. Further, gaming devices according to embodiments herein may be implemented using other computing devices and mobile devices, such as smart phones, tablets, and/or personal computers, among others.

FIG. 3A illustrates, in plain view, an example map **338** of a gaming area **340**. The gaming area **340** may, for example, be a casino floor. The map **338** shows the location of a plurality of EGMs **100** within the gaming area **340**. As will be appreciated, the locations of the EGMs **100** within a gaming area **340** are generally fixed, although EGMs may be relocated from time to time, such as when new EGMs are introduced, to create new traffic flow patterns within the gaming area **340**, to feature or highlight certain games, etc. As noted above, in order to assist the operation of the VR devices **200**, the VR/AR controller **114** may store a three dimensional wireframe map of the gaming area **340**, and may provide the three dimensional wireframe map to the VR viewers **200**.

An example of a wireframe map **342** for an SVE is shown in FIG. 3B. The wireframe map **342** is a three-dimensional model of the gaming area **340**. As shown in FIG. 3B, the wireframe map **342** includes wireframe models **344** that may correspond to the EGMs **100** that are physically in the gaming area **340**. The wireframe models **344** may also be entirely or partially virtual, e.g., existing only in the wireframe model **344** for the SVE. The wireframe models **344** may be pregenerated to correspond to various EGM form factors, such as single display EGMs, mechanical slot

EGMs, dual display EGMs, etc. The pregenerated models may then be placed into the wireframe map 342, for example, by a designer or other personnel. The wireframe map 342 may be updated at any time. For example, in an example where the wireframe map 342 corresponds to a real-world gaming area 340, the wireframe map 342 may be updated whenever the physical location of EGMs in the gaming area 340 is changed.

In some embodiments, the wireframe map 342 may be generated automatically using a VR/AR device 200, such as a 3D headset, that is configured to perform a three-dimensional depth scan of its surroundings and generate a three dimensional model based on the scan results. Thus, for example, an operator using a VR/AR device 200 may perform a walkthrough of the gaming area 340 while the VR/AR device 200 builds the 3D map of the gaming area.

The three dimensional wireframe map 342 may enable a VR/AR device 200 to more quickly and accurately determine its position and/or orientation within the gaming area. For example, a VR/AR device 200 may determine its location within the gaming area 340 using one or more position/orientation sensors. The VR/AR device 200 then builds a three dimensional map of its surroundings using depth scanning, and compares its sensed location relative to objects within the generated three dimensional map with an expected location based on the location of corresponding objects within the wireframe map 342. The VR/AR device 200 may calibrate or refine its position/orientation determination by comparing the sensed position of objects with the expected position of objects based on the wireframe map 342. Moreover, in an AR implementation, the VR/AR device 200 can be aware of objects or destinations within the gaming area 340 that it has not itself scanned, because the VR/AR device 200 has access to the wireframe map 342 of the entire gaming area 340. Processing requirements on the VR/AR device 200 may also be reduced because the wireframe map 342 is already available to the VR/AR device 200.

In some embodiments, the wireframe map 342 may store various information about EGMs in the gaming area, such as the identity, type, appearance, manufacturer, model, brand, color, texture, orientation and location of various types of EGMs, the locations of exits, bathrooms, courtesy desks, cashiers, ATMs, ticket redemption machines, etc. Such information may be used by a VR/AR device 200 to help the user navigate the gaming area. For example, if a user desires to find a destination within the gaming area, the user may ask the VR/AR device 200 for directions using a built-in microphone and voice recognition function in the VR/AR device 200 or use other hand gestures or eye/gaze controls tracked by the VR/AR device 200 (instead of or in addition to voice control). The VR/AR device 200 may process the request to identify the destination, and then may display a virtual object, such as a virtual path on the ground, virtual arrow, virtual sign, etc., to help the user to find the destination. In some embodiments, for example, the VR/AR device 200 may display a halo or glow around the destination to highlight it for the user, or have virtual 3D sounds coming from it so players could more easily find the machine.

According to some embodiments, a user of a VR/AR device 200 may use the VR/AR device 200 to obtain information about players and/or EGMs on a casino gaming floor. The information may be displayed to the user on the VR/AR device 200 in a number of different ways such as by displaying images on the VR/AR device 200 that appear to be three dimensional or two dimensional elements of the scene as viewed through the VR/AR device 200. In general,

the type and/or amount of data that is displayed to the user may depend on what type of user is using the VR/AR device 200 and, correspondingly, what level of permissions or access the user has. For example, a VR/AR device 200 may be operated in one of a number of modes, such as a player mode, an observer mode or an operator mode. In a player mode, the VR/AR device 200 may be used to display information about particular EGMs on a casino floor. The information may be generic information about an EGM or may be customized information about the EGM based on the identity or preferences of the user of the VR/AR device 200. In an observer mode, the VR/AR device 200 may be used to display information about particular EGMs on a casino floor or information about players of EGMs on the casino floor. In an operator mode, the VR/AR device 200 may also be used to display information about particular EGMs on a casino floor or information about players of EGMs on the casino floor, but the information may be different or more extensive than the information displayed to an observer.

FIGS. 4A-4C are views of a plurality of shared virtual environments (SVEs) 404A, 404B corresponding to fields of views (FOVs) of a plurality of players wearing head-mounted virtual reality devices 400A-400C, according to some embodiments. In this embodiment a real-time environmental model of a plurality of SVEs 404A, 404B is provided, with each SVE 404A, 404B including a plurality of virtual persons 406 and a plurality of virtual gaming devices 408.

As shown by FIG. 4A, first display data corresponding to a first SVE 404A is transmitted to a first player device 400A worn by a first player. The first display data includes user display data that causes a display device 402A in the first player device 400A to render a portion of the first SVE 404A based on a virtual orientation of the first player device 400A and a virtual location of the first player in the first SVE 404A. In the view shown by FIG. 4A, the portion of the SVE 404A is displayed as a 3D environment of a casino floor, with virtual persons 406, virtual gaming devices 408 arranged on the virtual casino floor. The virtual persons 406 may correspond to other real-world players, e.g., other players with their own VR devices, and/or system controlled players, e.g., virtual players that are controlled by a set of environmental rules, artificial intelligence, etc., to simulate a real-world casino environment. In some embodiments, the SVE 404A may also correspond to a real world environment, e.g., to a real-world casino floor with a similar or identical layout, and may include a full or partial floorplan corresponding to a floorplan of the real world environment, with elements such as virtual gaming devices 408 fully or partially corresponding to actual gaming device on the real-world casino floor. In other embodiments, the SVE 404A may be partially or entirely artificial, with little or no correspondence to a real world location, as desired.

As shown by FIG. 4B, second display data corresponding to a second SVE 404B, e.g., a different virtual casino, is transmitted to a second player device 400B worn by a second player. The second display data includes user display data that causes a display device 402B in the second player device 400B to render a portion of the second SVE 404B based on a virtual orientation of the second player device 400B and a virtual location of the second player in the second SVE 404B.

As with the SVE 404A above, the portion of the second SVE 404B in this example includes a plurality of virtual persons 406 and gaming devices 408. The virtual persons 406 may correspond to other real-world players and/or system controlled players, as desired. In this embodiment,

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the SVE 404B, which is different from the SVE 404A of FIG. 4A, includes a virtual first player 410B associated with the player interacting with the SVE 404A of FIG. 4A above.

In this manner, a player may interact with a primary SVE while appearing to other players in a different SVE at the same time. The player may simultaneously appear to other players in multiple SVEs, including the primary SVE. For example, as shown by FIG. 4C, third display data corresponding to the first SVE 400A is transmitted to a third player device 400C worn by a third player. The third display data includes user display data that causes a display device 402C in the third player device 400C to render a portion of the first SVE 404A based on a virtual orientation of the third player device 400C and a virtual location of the third player in the first SVE 400A.

In this embodiment, the first player appears as a virtual first player 410A in the first SVE 404A for the third player device 400C and as the virtual first player 410B in the first SVE 404A for the second player device 400B simultaneously, all while the first player interacts with the first SVE 404A.

In this example, the virtual person 410A appearing in the first SVE 404A may have a first appearance, and the virtual person 410B associated with the same player appearing in the second SVE 404B may have a second, different appearance. For example, the appearance of the virtual person 410A in the first SVE 404A may be based on a player preference of the third player, i.e., associated with the third player device 400C, as shown by FIG. 4C. For example, based on a player preference of the third player, the virtual person 410A may be rendered with particular apparel 412A, a perceived age, etc. Meanwhile, the appearance of the virtual person 410B in the second SVE 404B may be based on a player preference of the second player, which may cause the virtual person 410B associated with the same player to be entered with different apparel 412B, a different perceived age, etc. The virtual persons 410A, 410B may have different appearances on an SVE by SVE basis, e.g., with the virtual person having a common appearance in a particular SVE, and/or on a viewer by viewer basis, with the virtual person having a unique appearance for each viewing player based on individual player preferences or other criteria, including when the virtual player is being viewed in a common SVE by different players.

In some embodiments, a behavior of the virtual person 410A associated with the first player in the first SVE 404A may correspond to a behavior of the virtual person 410B associated with the first player in the second SVE 404B, and vice versa. For example, the first player may interact with elements of the first SVE 404A, such as other virtual persons 406 and/or gaming devices 408 in the first SVE 404A. These interactions may be partially or entirely duplicated and/or adapted for display within the second SVE 404B as well. For example, as shown by FIG. 4A, the player is interacting with a particular virtual gaming device 414A in the first SVE 404A. As shown by FIG. 4C, the virtual player 410A associated with the first player in the first SVE 404A is displayed to the third player as also interacting with the particular gaming device 414A in the first SVE 404A. At the same time, as shown by FIG. 4B, the virtual player 410B associated with the first player in the second SVE 404B is displayed to the second player as interacting with a corresponding particular virtual gaming device 414B in the second SVE 404B. The corresponding particular virtual gaming device 414B in the second SVE 404B may partially or entirely duplicate the appearance and/or features of the particular gaming device 414A in the first SVE 404A in

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some embodiments. In other embodiments, the particular virtual gaming device 414B in the second SVE 404B have a different appearance and/or functionality from the corresponding particular gaming device 414A in the first SVE 404A. For example, the first player may be playing a wagering game having a first theme 416A on the particular virtual gaming device 414A in the first SVE 404A, but the virtual player 410B in the second SVE 404B may appear to be playing a different wagering game having a different, second theme 416B at the corresponding particular virtual gaming device 414B in the second SVE 404B. In this embodiment as well, the virtual player 410A in the first SVE 404A may appear to the third player to be playing a different wagering game having a different, third theme 416C at the particular virtual gaming device 414A in the first SVE 404A, based on player preferences of the third player, for example, and/or based on other criteria.

As shown by FIGS. 5A and 5B, the SVEs 404A, 404B share a common floorplan 500 and layout with each other in this embodiment. For example, as discussed above, the floorplan 500 may correspond to a real-world casino environment, to attract players interested in visiting the real world casino, and who may already be familiar with the real world casino. In some embodiments, aspects of the real world casino may be modified in the SVEs 404A, 404B. For example, the layout and decor of the SVEs 404A, 404B may be modified to appear to be in a different time period, e.g., Las Vegas in the 1950s or 1970s, and may match the actual decor of the corresponding real world casino in that time period.

In some examples, the layouts may be the same across different SVEs 404A, 404B, but the individual virtual gaming devices 408 at particular locations may be different for different SVEs 404A, 404B, and/or for different players within the same SVE. For example, as shown by FIG. 5A, the first player may be playing at a particular virtual gaming device 414A in the first SVE 404A, with a first game type or theme, e.g., an Egyptian-themed slot game. Meanwhile, the second player, corresponding to virtual second player 418B in the second SVE 404B, may see the virtual first player 410B playing at the particular virtual gaming device 414B in the second SVE 404B, but with a different game theme, e.g., an old-west themed slot game, with the game activity and/or game results of the old-west themed slot game of the virtual gaming device 414B of the second SVE 404B corresponding to the actual game activity and/or game results of the Egyptian themed slot game of the virtual gaming device 414A being played by the first player in the first SVE 404A. Likewise, the third player may see the virtual first player 410A playing at the particular virtual gaming device 414A in the first SVE 404A with the same game theme, e.g., the Egyptian theme, or with another game theme, e.g., a licensed TV game show themed slot game. The third player may also be visible to the first player in the first SVE 404A as a virtual third player as well. In some examples, the game type may be modified for different SVEs and/or different players as well. For example, the first player may be playing a slot game at a particular virtual gaming device 414A in the first SVE 404A, and may simultaneously appear to be playing a video-poker game at the particular virtual gaming device 414B in the second SVE 404B, for example. In these and other examples, the behavior of the virtual first player 410B in the second SVE 404B may also be modified to correspond to the game and/or game type being displayed at the particular virtual gaming device 414B in the second SVE 404B, as desired.

These and other modifications may be applied to the SVEs **404A**, **404B** for all players, or may be modified on an SVE-by-SVE basis, with players in different SVEs experiencing different environmental elements and/or on a player-by-player basis, with different players in the same SVE experiencing different environmental elements, etc., as desired.

As discussed above, different SVEs may also having different floorplans and layouts and different subsets of virtual players, according to some embodiments. For example, as shown by FIGS. **6A** and **6B**, the first SVE **604A** of FIG. **6A** corresponds to a first virtual casino having a first floorplan **600A** with a first arrangement of virtual gaming devices **408**, while the second SVE **604B** of FIG. **6B** corresponds to a second virtual casino having a second floorplan **600B**, with a different arrangement of virtual gaming devices **408**. As noted above, the floorplans **600A**, **600B** and layouts of virtual gaming devices **408** of the SVEs **604A**, **604B** may correspond to floorplans and/or layouts of physical gaming devices in real-world casino environments, as desired.

In this example, a player playing at a particular virtual gaming device **614A** in the first SVE **604A** appears as a virtual first player **610A** playing at the virtual gaming device **614A** to a third player (i.e., virtual third player **620A**) in the first SVE **604A** and simultaneously appears as a virtual first player **410B** playing at a corresponding virtual gaming device **614B**, which may be at a different relative location in the floorplan **600B** of the second SVE **604B**, to a second player (i.e., virtual second player **618B**) in the second SVE **604B**.

In some examples, virtual persons associated with players may enter and exit different SVEs based on system and/or player virtual occupancy preferences, and/or other criteria. In this regard, FIGS. **7A-7C** are floorplan views of an SVE **704** showing addition and removal of virtual players based on virtual occupancy preferences and rules, according to some embodiments. As shown by FIG. **7A**, a player playing in another SVE (not shown) appears to a second player (i.e., virtual second player **718**) as a virtual first player **710** playing at a particular virtual gaming device **714** of a plurality of virtual gaming devices **408** in the SVE **704**. In this example, the second player has an associated occupancy preference that include a preferred number and/or range of numbers of virtual persons in the SVE **704**.

As shown by FIG. **7B**, virtual persons **406**, which may correspond to real-world players and/or system controlled virtual players, may enter and exit the SVE **704** over time. If the number of virtual persons **406** varies from the preferred number of virtual persons for the occupancy preference of the second player, the system may cause one or more virtual persons **406** to enter or exit the SVE **704**. For example, the system may determine that a virtual third player **722** associated with a third player has entered the SVE **704**. Based on the occupancy preference of the second player, the system may then cause another virtual person to leave the SVE **704** to maintain the number of virtual persons **406** in the SVE at the preferred number of virtual persons **406**.

For example, as shown by FIG. **7C**, the system may determine that the virtual first player **710** should exit the SVE **704**, and may take control of the virtual first player **710** to cause the virtual first player to exit the SVE **704**. Meanwhile, the first player may continue playing in the other SVE without interruption, and in some embodiments, without being notified of the change to the SVE **704**.

In some embodiments, a player may be able to move between SVEs while maintaining the corresponding virtual player in both SVEs without interruption. For example, a player input by the player may indicate a desire to interact with a different SVE, which may in turn cause the player device to stop rendering the current SVE and begin rendering the different SVE. In some embodiments, this can be implemented based on an SVE preference for the player or other criteria, such as virtual occupancy of different SVEs for example, as desired.

FIG. **8** is a flowchart illustrating operations **800** of systems/methods for facilitating embodiments described herein. The operations **800** may be performed by one or more processor circuits of one or more computing devices, such as any of the computing devices described herein, for example. The operations **800** may include providing, by a processor circuit, a real-time environmental model of a plurality of shared virtual environments (SVEs) (Block **802**). Each SVE may include a plurality of virtual persons and a plurality of virtual gaming devices. For example, using the embodiment of FIGS. **4A-4C** above, a real time model including the different SVEs **404A**, **404B** may be provided.

The operations **800** may further include transmitting first display data corresponding to a first SVE of the plurality of SVEs to a first player device worn by a first player, including user display data that causes a display device in the first player device to render a portion of the first SVE based on a virtual orientation of the first player device and a virtual location of the first player in the first SVE (Block **804**). Referring again to the embodiment of FIGS. **4A-4C** above, FIG. **4A** shows an example of a display device **402A** of a first player's player device **400A** rendering a portion of the first SVE **404A** based on virtual orientation and location in the SVE **404A**.

The operations **800** may further include transmitting second display data corresponding to a second SVE of the plurality of SVEs to a second player device worn by a second player, including user display data that causes a display device in the second player device to render a portion of the second SVE based on a virtual orientation of the second player device and a virtual location of the second player in the second SVE, the portion of the second SVE comprises a virtual person associated with the first player (Block **806**). Referring again to the embodiment of FIGS. **4A-4C** above, FIG. **4B** shows an example of a display device **402B** of a second player's player device **400B** rendering a portion of the second SVE **404B** based on virtual orientation and location in the SVE **404B**, including a virtual person **410B** associated with the first player appearing in the second SVE **404B**.

Embodiments described herein may be implemented in various configurations for gaming devices **100**, including but not limited to: (1) a dedicated gaming device, wherein the computerized instructions for controlling any games (which are provided by the gaming device) are provided with the gaming device prior to delivery to a gaming establishment; and (2) a changeable gaming device, where the computerized instructions for controlling any games (which are provided by the gaming device) are downloadable to the gaming device through a data network when the gaming device is in a gaming establishment. In some embodiments, the computerized instructions for controlling any games are executed by at least one central server, central controller or remote host. In such a "thin client" embodiment, the central server remotely controls any games (or other suitable interfaces) and the gaming device is utilized to display such games (or suitable interfaces) and receive one

or more inputs or commands from a player. In another embodiment, the computerized instructions for controlling any games are communicated from the central server, central controller or remote host to a gaming device local processor and memory devices. In such a “thick client” embodiment, the gaming device local processor executes the communicated computerized instructions to control any games (or other suitable interfaces) provided to a player.

In some embodiments, a gaming device may be operated by a mobile device, such as a mobile telephone, tablet or other mobile computing device. For example, a mobile device may be communicatively coupled to a gaming device and may include a user interface that receives user inputs that are received to control the gaming device. The user inputs may be received by the gaming device via the mobile device.

In some embodiments, one or more gaming devices in a gaming system may be thin client gaming devices and one or more gaming devices in the gaming system may be thick client gaming devices. In another embodiment, certain functions of the gaming device are implemented in a thin client environment and certain other functions of the gaming device are implemented in a thick client environment. In one such embodiment, computerized instructions for controlling any primary games are communicated from the central server to the gaming device in a thick client configuration and computerized instructions for controlling any secondary games or bonus functions are executed by a central server in a thin client configuration.

The present disclosure contemplates a variety of different gaming systems each having one or more of a plurality of different features, attributes, or characteristics. It should be appreciated that a “gaming system” as used herein refers to various configurations of: (a) one or more central servers, central controllers, or remote hosts; (b) one or more gaming devices; and/or (c) one or more personal gaming devices, such as desktop computers, laptop computers, tablet computers or computing devices, PDAs, mobile telephones such as smart phones, and other mobile computing devices.

In certain such embodiments, computerized instructions for controlling any games (such as any primary or base games and/or any secondary or bonus games) displayed by the gaming device are executed by the central server, central controller, or remote host. In such “thin client” embodiments, the central server, central controller, or remote host remotely controls any games (or other suitable interfaces) displayed by the gaming device, and the gaming device is utilized to display such games (or suitable interfaces) and to receive one or more inputs or commands. In other such embodiments, computerized instructions for controlling any games displayed by the gaming device are communicated from the central server, central controller, or remote host to the gaming device and are stored in at least one memory device of the gaming device. In such “thick client” embodiments, the at least one processor of the gaming device executes the computerized instructions to control any games (or other suitable interfaces) displayed by the gaming device.

In some embodiments in which the gaming system includes: (a) a gaming device configured to communicate with a central server, central controller, or remote host through a data network; and/or (b) a plurality of gaming devices configured to communicate with one another through a data network, the data network is an internet or an intranet. In certain such embodiments, an internet browser of the gaming device is usable to access an internet game page from any location where an internet connection is available. In one such embodiment, after the internet game page is

accessed, the central server, central controller, or remote host identifies a player prior to enabling that player to place any wagers on any plays of any wagering games. In one example, the central server, central controller, or remote host identifies the player by requiring a player account of the player to be logged into via an input of a unique username and password combination assigned to the player. It should be appreciated, however, that the central server, central controller, or remote host may identify the player in any other suitable manner, such as by validating a player tracking identification number associated with the player; by reading a player tracking card or other smart card inserted into a card reader (as described below); by validating a unique player identification number associated with the player by the central server, central controller, or remote host; or by identifying the gaming device, such as by identifying the MAC address or the IP address of the internet facilitator. In various embodiments, once the central server, central controller, or remote host identifies the player, the central server, central controller, or remote host enables placement of one or more wagers on one or more plays of one or more primary or base games and/or one or more secondary or bonus games, and displays those plays via the internet browser of the gaming device.

It should be appreciated that the central server, central controller, or remote host and the gaming device are configured to connect to the data network or remote communications link in any suitable manner. In various embodiments, such a connection is accomplished via: a conventional phone line or other data transmission line, a digital subscriber line (DSL), a T-1 line, a coaxial cable, a fiber optic cable, a wireless or wired routing device, a mobile communications network connection (such as a cellular network or mobile internet network), or any other suitable medium. It should be appreciated that the expansion in the quantity of computing devices and the quantity and speed of internet connections in recent years increases opportunities for players to use a variety of gaming devices to play games from an ever-increasing quantity of remote sites. It should also be appreciated that the enhanced bandwidth of digital wireless communications may render such technology suitable for some or all communications, particularly if such communications are encrypted. Higher data transmission speeds may be useful for enhancing the sophistication and response of the display and interaction with players.

In the above-description of various embodiments, various aspects may be illustrated and described herein in any of a number of patentable classes or contexts including any new and useful process, machine, manufacture, or composition of matter, or any new and useful improvement thereof. Accordingly, various embodiments described herein may be implemented entirely by hardware, entirely by software (including firmware, resident software, micro-code, etc.) or by combining software and hardware implementation that may all generally be referred to herein as a “circuit,” “module,” “component,” or “system.” Furthermore, various embodiments described herein may take the form of a computer program product including one or more computer readable media having computer readable program code embodied thereon.

Any combination of one or more computer readable media may be used. The computer readable media may be a computer readable signal medium or a computer readable storage medium. A computer readable storage medium may be, for example, but not limited to, an electronic, magnetic, optical, electromagnetic, or semiconductor system, apparatus, or device, or any suitable combination of the foregoing.

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More specific examples (a non-exhaustive list) of the computer readable storage medium would include the following: a portable computer diskette, a hard disk, a random access memory (RAM), a read-only memory (ROM), an erasable programmable read-only memory (EPROM or Flash memory), an appropriate optical fiber with a repeater, a portable compact disc read-only memory (CD-ROM), an optical storage device, a magnetic storage device, or any suitable combination of the foregoing. In the context of this document, a computer readable storage medium may be any medium that can contain, or store a program for use by or in connection with an instruction execution system, apparatus, or device.

A computer readable signal medium may include a propagated data signal with computer readable program code embodied therein, for example, in baseband or as part of a carrier wave. Such a propagated signal may take any of a variety of forms, including, but not limited to, electromagnetic, optical, or any suitable combination thereof. A computer readable signal medium may be any computer readable medium that is not a computer readable storage medium and that can communicate, propagate, or transport a program for use by or in connection with an instruction execution system, apparatus, or device. Program code embodied on a computer readable signal medium may be transmitted using any appropriate medium, including but not limited to wireless, wireline, optical fiber cable, radio frequency ("RF"), etc., or any suitable combination of the foregoing.

Computer program code for carrying out operations for aspects of the present disclosure may be written in any combination of one or more programming languages, including an object oriented programming language such as Java, Scala, Smalltalk, Eiffel, JADE, Emerald, C++, C #, VB.NET, Python or the like, conventional procedural programming languages, such as the "C" programming language, Visual Basic, Fortran 2003, Perl, Common Business Oriented Language ("COBOL") 2002, PHP: Hypertext Processor ("PHP"), Advanced Business Application Programming ("ABAP"), dynamic programming languages such as Python, Ruby and Groovy, or other programming languages. The program code may execute entirely on the user's computer, partly on the user's computer, as a stand-alone software package, partly on the user's computer and partly on a remote computer or entirely on the remote computer or server. In the latter scenario, the remote computer may be connected to the user's computer through any type of network, including a local area network (LAN) or a wide area network (WAN), or the connection may be made to an external computer (for example, through the Internet using an Internet Service Provider) or in a cloud computing environment or offered as a service such as a Software as a Service (SaaS).

Various embodiments were described herein with reference to flowchart illustrations and/or block diagrams of methods, apparatus (systems), devices and computer program products according to various embodiments described herein. It will be understood that each block of the flowchart illustrations and/or block diagrams, and combinations of blocks in the flowchart illustrations and/or block diagrams, can be implemented by computer program instructions. These computer program instructions may be provided to a processing circuit of a general purpose computer, special purpose computer, or other programmable data processing apparatus to produce a machine, such that the instructions, which execute via the processing circuit of the computer or other programmable instruction execution apparatus, create

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a mechanism for implementing the functions/acts specified in the flowchart and/or block diagram block or blocks.

These computer program instructions may also be stored in a computer readable medium that when executed can direct a computer, other programmable data processing apparatus, or other devices to function in a particular manner, such that the instructions when stored in the computer readable medium produce an article of manufacture including instructions which when executed, cause a computer to implement the function/act specified in the flowchart and/or block diagram block or blocks. The computer program instructions may also be loaded onto a computer, other programmable instruction execution apparatus, or other devices to cause a series of operations to be performed on the computer, other programmable apparatuses or other devices to produce a computer implemented process such that the instructions which execute on the computer or other programmable apparatus provide processes for implementing the functions/acts specified in the flowchart and/or block diagram block or blocks.

The flowchart and block diagrams in the figures illustrate the architecture, functionality, and operation of possible implementations of systems, methods, and computer program products according to various aspects of the present disclosure. In this regard, each block in the flowchart or block diagrams may represent a module, segment, or portion of code, which includes one or more executable instructions for implementing the specified logical function(s). It should also be noted that, in some alternative implementations, the functions noted in the block may occur out of the order noted in the figures. For example, two blocks shown in succession may, in fact, be executed substantially concurrently, or the blocks may sometimes be executed in the reverse order, depending upon the functionality involved. It will also be noted that each block of the block diagrams and/or flowchart illustration, and combinations of blocks in the block diagrams and/or flowchart illustration, can be implemented by special purpose hardware-based systems that perform the specified functions or acts, or combinations of special purpose hardware and computer instructions.

The terminology used herein is for the purpose of describing particular aspects only and is not intended to be limiting of the disclosure. As used herein, the singular forms "a", "an" and "the" are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms "comprises" and/or "comprising," when used in this specification, specify the presence of stated features, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, steps, operations, elements, components, and/or groups thereof. As used herein, the term "and/or" includes any and all combinations of one or more of the associated listed items and may be designated as "/". Like reference numbers signify like elements throughout the description of the figures.

Many different embodiments have been disclosed herein, in connection with the above description and the drawings. It will be understood that it would be unduly repetitious and obfuscating to literally describe and illustrate every combination and subcombination of these embodiments. Accordingly, all embodiments can be combined in any way and/or combination, and the present specification, including the drawings, shall be construed to constitute a complete written description of all combinations and subcombinations of the embodiments described herein, and of the manner and process of making and using them, and shall support claims to any such combination or subcombination.

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What is claimed is:

1. A system comprising:  
a processor circuit; and  
a memory comprising machine-readable instructions that,  
when executed by the processor circuit, cause the  
processor circuit to:  
provide a real-time environmental model of a plurality of  
shared virtual environments (SVEs), each SVE comprising a plurality of virtual persons and a plurality of  
virtual gaming devices;  
transmit first display data corresponding to a first SVE of  
the plurality of SVEs to a first player device worn by a  
first player, the first display data comprising user display  
data that causes a display device in the first player  
device to render a portion of the first SVE based on a  
virtual orientation of the first player device and a virtual  
location of the first player in the first SVE;  
transmit second display data corresponding to a second  
SVE of the plurality of SVEs to a second player device  
worn by a second player, the second display data  
comprising user display data that causes a display  
device in the second player device to render a portion  
of the second SVE based on a virtual orientation of the  
second player device and a virtual location of the  
second player in the second SVE, wherein the portion  
of the second SVE comprises a virtual person associated  
with the first player;  
receive a player input via an input device of the first  
player device; and  
based on the player input, cause the display device of the  
first player device to:  
receive third display data corresponding to the second  
SVE, the third display data comprising user display  
data that causes the display device in the first player  
device to render a portion of the second SVE based  
on a virtual orientation of the first player device and  
a virtual location of the first player in the second  
SVE;  
stop rendering the portion of the first SVE; and  
render the portion of the second SVE.
2. The system of claim 1, wherein the instructions further  
cause the processor circuit to:  
transmit fourth display data corresponding to the first  
SVE to a third player device worn by a third player, the  
fourth display data comprising user display data that  
causes a display device in the third player device to  
render a portion of the first SVE based on a virtual  
orientation of the third player device and a virtual  
location of the third player in the first SVE, wherein the  
portion of the first SVE comprises a virtual person  
associated with the first player.
3. The system of claim 2, wherein a first appearance of the  
virtual person associated with the first player in the second  
SVE is based on a player preference of the second player,  
and wherein a second appearance of the virtual person  
associated with the first player in the first SVE is based on  
a player preference of the third player.
4. The system of claim 1, wherein a behavior of the virtual  
person associated with the first player in the second SVE  
corresponds to a behavior of virtual person associated with  
the first player in the first SVE.
5. The system of claim 1, wherein a floorplan of the first  
SVE is different from a floorplan of the second SVE.
6. The system of claim 1, wherein the plurality of virtual  
gaming devices of the first SVE is different from the  
plurality of virtual gaming devices of the second SVE.

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7. The system of claim 1, wherein the plurality of virtual  
gaming devices of the first SVE correspond to a plurality of  
physical gaming devices in a casino environment.

8. The system of claim 1, wherein a player preference  
associated with the second player comprises an occupancy  
preference comprising a preferred number of virtual persons  
in the second SVE, and

wherein instructions further cause the processor circuit to;  
determine that a virtual person associated with a third  
player has entered the second SVE; and  
based on the occupancy preference of the second  
player, cause a virtual person to leave the SVE to  
maintain the number of virtual persons in the SVE at  
the preferred number of virtual persons.

9. The system of claim 1, wherein the instructions further  
cause the processor circuit to:

determine a SVE preference for the first player; and  
based on the SVE preference, cause the display device of  
the first player device to selectively render one of the  
portion of the first SVE and the portion of the second  
SVE.

10. A method comprising:

providing, by a processor circuit, a real-time environmen-  
tal model of a plurality of shared virtual environments  
(SVEs), each SVE comprising a plurality of virtual  
persons and a plurality of virtual gaming devices;

transmitting first display data corresponding to a first SVE  
of the plurality of SVEs to a first player device worn by  
a first player, the first display data comprising user  
display data that causes a display device in the first  
player device to render a portion of the first SVE based  
on a virtual orientation of the first player device and a  
virtual location of the first player in the first SVE; and  
transmitting second display data corresponding to a second  
SVE of the plurality of SVEs to the first player  
device, the second display data comprising user display  
data that causes the display device in the first player  
device to render a portion of the second SVE based on  
a virtual orientation of the first player device and a  
virtual location of the first player in the second SVE;  
receiving a player input via an input device of the first  
player device; and

based on the player input, cause the display device of the  
first player device to:  
stop rendering the portion of the first SVE; and  
render the portion of the second SVE.

11. The method of claim 10, further comprising:

transmitting third display data corresponding to the first  
SVE to a third player device worn by a third player, the  
third display data comprising user display data that  
causes a display device in the third player device to  
render a portion of the first SVE based on a virtual  
orientation of the third player device and a virtual  
location of the third player in the first SVE, wherein the  
portion of the first SVE comprises a virtual person  
associated with the first player.

12. The method of claim 11, wherein a first appearance of  
the virtual person associated with the first player in the  
second SVE is based on a player preference of the second  
player, and wherein a second appearance of the virtual  
person associated with the first player in the first SVE is  
based on a player preference of the third player.

13. The method of claim 11, wherein a behavior of the  
virtual person associated with the first player in the second  
SVE corresponds to a behavior of the first player in the first  
SVE.

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14. The method of claim 11, wherein a floorplan of the first SVE is different from a floorplan of the second SVE.

15. The method of claim 11, wherein the plurality of virtual gaming devices of the first SVE is different from the plurality of virtual gaming devices of the second SVE. 5

16. The method of claim 11, wherein the plurality of virtual gaming devices of the first SVE correspond to a plurality of physical gaming devices in a casino environment.

17. A virtual-reality player device comprising: 10

a head-wearable frame;

an input device;

a display device coupled to the frame to position the display device in a field of view of a first player; 15

a processor circuit; and

a memory comprising machine-readable instructions that, when executed by the processor circuit, cause the processor circuit to:

receive first display data corresponding to a first SVE 20

of a plurality of SVEs, each SVE comprising a plurality of virtual persons and a plurality of virtual gaming devices;

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render, by the display device, a portion of the first SVE based on a virtual orientation of the player device and a virtual location of the first player in the first SVE; and

receive second display data corresponding to a second SVE of the plurality of SVEs;

receive a player input via the input device; and based on the player input:

stop rendering the portion of the first SVE; and

render a portion of the second SVE based on a virtual orientation of the player device and a virtual location of the first player in the second SVE.

18. The player device of claim 17, wherein the instructions further cause the processor circuit to:

transmit third display data corresponding to the first SVE to a second player device worn by a second player, the second display data comprising user display data that causes a display device in the second player device to render a portion of the first SVE based on a virtual orientation of the second player device and a virtual location of the second player in the first SVE, wherein the portion of the first SVE comprises a virtual person associated with the first player.

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